

Codex GPT-5.4 Reference vs Claude Code Qwen2.5-Coder 32B Q4_K_M Evaluation

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Goal

This report evaluates the Claude Code benchmark answer run in ``docs/evaluations/answers/claude_code_qwen25_32b_q4km_20260619T112026Z.json`` against the Codex GPT-5.4 reference answers stored in ``docs/evaluations/eval_answers_v2.json``. It explains, question by question, whether the Claude Code answer agrees with the reference answer, partially covers it, or misses the core expected answer.

Test Environment

Codex Reference Side

- Model: ``GPT-5.4 (Codex)``
- Reference file: ``docs/evaluations/eval_answers_v2.json``
- Question source: ``eval_questions_v2.json``
- Evaluation basis: semantic comparison against the Codex reference answer for each question

Claude Code / Ollama Side

- Claude Code model id: ``claude-qwen25-32b-q4km``
- Run label: ``qwen25_32b_q4km``
- Ollama tag: ``qwen2.5-coder:32b-instruct-q4_K_M``
- Host: ``merlin-g-100.psi.ch``
- Anthropic-compatible gateway: ``http://127.0.0.1:4001``
- Allowed Claude Code tools: ``Read,Grep,Glob``
- Run started: ``2026-06-19T11:20:26.153341+00:00``
- Run finished: ``2026-06-19T12:28:25.571628+00:00``
- Recorded answer count: ``100``
- Recorded failure count in metadata: ``0``

Important Caveat

The verdicts are a structured comparison against the Codex reference answers, not a fresh manual audit of every IPPL source file. Correct means the Claude Code answer captures the central reference facts; Partial means it contains relevant signal but misses important scope, exact location, or implementation detail; Incorrect means it failed, abstained, pointed to the wrong subsystem, or omitted the core reference answer.

Because Claude Code answers are often much longer than the concise reference answers, the comparison emphasizes central technical nouns, exact identifiers, files, classes, functions, and implementation steps rather than exact wording.

Overall Result

Metric Value
--- ---:
Questions 100
Correct 33
Partial 46
Incorrect 21
Average score (<code>`Correct=1`</code> , <code>`Partial=0.5`</code> , <code>`Incorrect=0`</code>) 0.560
Runtime/API failures counted as incorrect 0
Answers with leaked self-correction/planning text 95
Mean answer latency 40.79s
Median answer latency 32.18s

Main Findings

- Strongest areas: Numerical Meaning, Testing And Workflow. These categories mostly reward direct repository navigation, API identification, and concise file/class role descriptions.
- Weakest areas: Build And Install, Definition Location. These categories were hurt by missing exact implementation locations, shallow source inspection, or cross-file reasoning gaps.
- The run produced ``0`` runtime/API failures, so most lost points came from answer quality rather than crashed generations.
- The Claude Code/Ollama setup had mean latency ``40.79s`` per question, with a median of ``32.18s``.
- ``95`` answers were flagged for leaked planning/self-correction text, such as step-by-step source-inspection narration.

Category Breakdown

Category Questions Correct Partial Incorrect Avg score
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	13	3	4	6	0.385
File Purpose	13	3	4	6	0.385
Definition Location	15	0	11	4	0.367
Class Responsibility	13	7	5	1	0.731
Algorithm	12	5	3	4	0.542
Data Flow	9	2	6	1	0.556
Api Usage	12	6	3	3	0.625
Parallelism And Kokkos	9	4	5	0	0.722
Boundary And Halo	5	1	4	0	0.600
Numerical Meaning	3	3	0	0	1.000
Build And Install	5	0	3	2	0.300
Testing And Workflow	4	2	2	0	0.750

Detailed Per-Question Evaluation

File Purpose

file_001 - What does Ippl.h do?

- Verdict: ****Partial****
- Correct reference answer: Declares the top-level IPPL runtime handles (``Comm``, ``Env``, ``Info``, ``Warn``, ``Error``) and the lifecycle wrappers ``initialize``, ``finalize``, ``fence``, and ``abort``.
- Explanation: The answer overlaps with the Codex reference, but misses important scope, implementation detail, or exact terminology.
- Reference-keyword coverage: ``0.00``
- Reference identifier/location hits: ``1/9``
- Latency: ``45.27s``
- Matched reference signals: ``Comm``
- Missing/weak reference signals: ``top-level, runtime, handles, comm, env, info, warn, error``
- Claude Code answer excerpt: ``To determine what `Ippl.h` does in the IPPL (Integrated Parallel Primitives Library) source tree, I'll explore its contents and any associated documentation. Given that the source tree is located at `data/raw/ippl`, I will read the file there. ### Step 1: Read the contents of `Ippl.h` Let's start by reading the `Ipp...``

file_002 - What is the purpose of Ippl.cpp?

- Verdict: ****Incorrect****
- Correct reference answer: Implements runtime setup/teardown: command-line option parsing, MPI communicator/environment creation, logger setup, Kokkos init/finalize, and abort/fence helpers.
- Explanation: The answer does not recover the main Codex reference point or points to insufficiently related implementation details.
- Reference-keyword coverage: ``0.13``
- Latency: ``31.21s``
- Matched reference signals: ``mpi, setup``
- Missing/weak reference signals: ``implements, runtime, setup/teardown, command-line, option, parsing, communicator/environment, creation``
- Claude Code answer excerpt: ``To determine the purpose of `Ippl.cpp`, I'll explore the source code and documentation within the IPPL (Integrated Physics and Parallelism Library) repository under `data/raw/ippl`. ### Step 1: Read `Ippl.cpp` Let's start by examining the contents of `Ippl.cpp` to understand its primary functions and components. `````

file_003 - What does FFT.hpp implement?

- Verdict: ****Incorrect****
- Correct reference answer: Implements ``FFTBase`` setup plus the FFT specialization family for complex/real, sine, cosine, and cosine-I transforms on IPPL fields via `heFFTe`.
- Explanation: The answer does not recover the main Codex reference point or points to insufficiently related implementation details.
- Reference-keyword coverage: ``0.15``
- Reference identifier/location hits: ``0/1``
- Latency: ``20.20s``
- Matched reference signals: ``implements, fft``
- Missing/weak reference signals: ``fftbases, setup, specialization, family, complex/real, sine, cosine, cosine-i``
- Claude Code answer excerpt: ``To determine what `FFT.hpp` implements, I'll use the read-only tools to explore the contents of the file located at `data/raw/ippl`. Let's start by reading the file and identifying its key components. ### Step 1: Read the FFT.hpp File ``shell read_file data/raw/ippl/FFT.hpp `` <system-reminder> Here is the content...``

file_004 - What is BareField.h responsible for?

- Verdict: ****Partial****
- Correct reference answer: Defines the core distributed field container: Kokkos-backed storage, owned/allocated domains, field layout access, halo exchange hooks, and expression-style assignment.

- Explanation: The answer overlaps with the Codex reference, but misses important scope, implementation detail, or exact terminology.
- Reference-keyword coverage: `0.19`
- Latency: `26.15s`
- Matched reference signals: `field, storage`
- Missing/weak reference signals: `core, distributed, container, kokkos-backed, owned/allocated, domains, layout, access`
- Claude Code answer excerpt: `To determine the purpose of `BareField.h`, I'll search through the IPPL source tree located in `data/raw/ippl` and look for relevant comments, class definitions, or usage within the codebase. Let's start by searching for `BareField.h` and understanding its contents. ### Step 1: Locate `BareField.h` First, we need to...`

file_005 - What does Field.h add on top of BareField?

- Verdict: ****Correct****
- Correct reference answer: Adds mesh awareness and boundary-condition management on top of `BareField`, plus volume integral/average helpers.
- Explanation: The answer captures the central facts of the Codex reference and includes the important technical nouns, symbols, or implementation locations.
- Reference-keyword coverage: `0.25`
- Reference identifier/location hits: `1/1`
- Latency: `93.13s`
- Matched reference signals: `management, barefield, BareField`
- Missing/weak reference signals: `mesh, awareness, boundary-condition, volume, integral/average, helpers`
- Claude Code answer excerpt: `To answer the question about what `Field.h` adds on top of `BareField`, I'll need to explore the contents of these files within the IPPL source tree located at `data/raw/ippl`. Let's start by reading both `Field.h` and `BareField.h` to understand their contents and relationships. Read file: `data/raw/ippl/Field.h` R...`

file_006 - What role does HaloCells.h play in IPPL?

- Verdict: ****Correct****
- Correct reference answer: Defines the halo-exchange machinery for fields: pack/unpack, boundary exchange, and periodic handling for ghost cells.
- Explanation: The answer captures the central facts of the Codex reference and includes the important technical nouns, symbols, or implementation locations.
- Reference-keyword coverage: `0.40`
- Latency: `52.84s`
- Matched reference signals: `boundary, exchange, ghost, cells`
- Missing/weak reference signals: `halo-exchange, machinery, fields, pack/unpack, periodic, handling`
- Claude Code answer excerpt: `To determine the role of `HaloCells.h` in IPPL, I'll explore the source files under `data/raw/ippl` to find relevant information about this header file. Let's start by reading the contents of `HaloCells.h` and see if there are any comments or declarations that give insight into its purpose. ```bash read_file data/ra...`

file_007 - What is the purpose of FieldLayout.h?

- Verdict: ****Incorrect****
- Correct reference answer: Defines how a global `NDIndex` domain is partitioned across MPI ranks, including local subdomains, neighbors, and communication ranges.
- Explanation: The answer does not recover the main Codex reference point or points to insufficiently related implementation details.
- Reference-keyword coverage: `0.17`
- Reference identifier/location hits: `0/1`
- Latency: `34.45s`
- Matched reference signals: `domain, across`
- Missing/weak reference signals: `global, ndindex, partitioned, mpi, ranks, local, subdomains, neighbors`
- Claude Code answer excerpt: `To determine the purpose of `FieldLayout.h`, I'll explore its content and usage within the IPPL (Integrated Particle Physics Library) source tree located at `data/raw/ippl`. ### Step 1: Read the Header File First, let's read the contents of `FieldLayout.h` to understand what it defines and declares. \*\*Tool Call:\*\* R...`

file_008 - What does ParticleBase.h define?

- Verdict: ****Partial****
- Correct reference answer: Defines the generic particle container base class with positions, optional IDs, registered attributes, and a particle layout that controls migration.
- Explanation: The answer overlaps with the Codex reference, but misses important scope, implementation detail, or exact terminology.
- Reference-keyword coverage: `0.31`
- Latency: `33.89s`
- Matched reference signals: `particle, container, base`
- Missing/weak reference signals: `generic, positions, optional, ids, registered, attributes, layout, controls`

- Claude Code answer excerpt: `To determine what `ParticleBase.h` defines, I'll use the `Read` tool to examine its contents. Read `/psi/home/ext-nobile_a/rag-scientific-code-agent/data/raw/ippl/src/base/ParticleBase.h` ---
`\*\*File:\*\* `data/raw/ippl/src/base/ParticleBase.h` ``cpp #ifndef PARTICLEBASE_H #define PARTICLEBASE_H #include <vector> #inc...`

file_009 - What is the purpose of ParticleSpatialLayout.h?

- Verdict: ****Partial****
- Correct reference answer: Defines the spatial particle layout that decides ownership by region and migrates particles to the correct rank after motion.
- Explanation: The answer overlaps with the Codex reference, but misses important scope, implementation detail, or exact terminology.
- Reference-keyword coverage: `0.33`
- Latency: `42.40s`
- Matched reference signals: `spatial, particle, layout, particles`
- Missing/weak reference signals: `decides, ownership, region, migrates, correct, rank, after, motion`
- Claude Code answer excerpt: `To determine the purpose of `ParticleSpatialLayout.h`, I'll explore the relevant files in the IPPL (Innovative Particle Physics Laboratory) source tree under `data/raw/ippl`. ### Step 1: Locate `ParticleSpatialLayout.h` First, I'll check if `ParticleSpatialLayout.h` exists and where it is located.
``bash find data/...`

file_010 - What does ParticleAttrib.h implement?

- Verdict: ****Correct****
- Correct reference answer: Implements the `ParticleAttrib` storage object: a Kokkos-view-backed particle attribute with resize, serialization, and mirror/access helpers.
- Explanation: The answer captures the central facts of the Codex reference and includes the important technical nouns, symbols, or implementation locations.
- Reference-keyword coverage: `0.27`
- Reference identifier/location hits: `1/1`
- Latency: `35.09s`
- Matched reference signals: `implements, particle, attribute, ParticleAttrib`
- Missing/weak reference signals: `particleattrib, storage, object, kokkos-view-backed, resize, serialization, mirror/access, helpers`
- Claude Code answer excerpt: `To determine what `ParticleAttrib.h` implements, I'll use the read-only tools to explore the source files under `data/raw/ippl`. Let's start by reading `ParticleAttrib.h` directly. Read: data/raw/ippl/include/ParticleAttrib.h ``cpp // ParticleAttrib.h // // Purpose: Define classes and structures for particle attrib...`

file_011 - What is UniformCartesian.h for?

- Verdict: ****Incorrect****
- Correct reference answer: Defines the uniform Cartesian mesh class with spacing, origin, cell volume, total mesh volume, and vertex-position helpers.
- Explanation: The answer does not recover the main Codex reference point or points to insufficiently related implementation details.
- Reference-keyword coverage: `0.17`
- Latency: `21.75s`
- Matched reference signals: `uniform, cartesian`
- Missing/weak reference signals: `mesh, spacing, origin, cell, volume, total, vertex-position, helpers`
- Claude Code answer excerpt: `To determine the purpose of `UniformCartesian.h`, I'll explore the source code in `data/raw/ippl` and look for any relevant documentation or comments. Let's start by searching for `UniformCartesian.h` within the repository to see if there are any files that include it or reference its usage:
``bash find data/raw/ip...`

file_012 - What does OrthogonalRecursiveBisection.h do?

- Verdict: ****Incorrect****
- Correct reference answer: Declares the orthogonal recursive bisection domain-decomposition/load-balancing class and its cut/median/repartition helpers.
- Explanation: The answer does not recover the main Codex reference point or points to insufficiently related implementation details.
- Reference-keyword coverage: `0.00`
- Latency: `8.72s`
- Missing/weak reference signals: `orthogonal, recursive, bisection, domain-decomposition/load-balancing, cut/median/repartition, helpers`
- Claude Code answer excerpt: `To determine the purpose of `OrthogonalRecursiveBisection.h`, I'll read through the relevant parts of this header file and any associated implementation files. This will help me understand the functionality and roles defined within it. ### Step 1: Read `OrthogonalRecursiveBisection.h` I'll start by reading the conte...`

file_013 - What is the purpose of CIC.h in the Interpolation subsystem?

- Verdict: ****Incorrect****
- Correct reference answer: Declares the first-order cloud-in-cell interpolation helpers for scatter/gather between particles and grid fields.
- Explanation: The answer does not recover the main Codex reference point or points to insufficiently related implementation details.
- Reference-keyword coverage: `0.11`
- Latency: `21.13s`
- Matched reference signals: `interpolation`
- Missing/weak reference signals: `first-order, cloud-in-cell, helpers, scatter/gather, between, particles, grid, fields`
- Claude Code answer excerpt: `To determine the purpose of `CIC.h` in the Interpolation subsystem, I'll explore the relevant files and symbols within the `data/raw/ippl` directory. **### Step 1: Locate the Interpolation Subsystem First, let's find the Interpolation subsystem in the IPPL source tree. **\*\*Tool Call:\*\*** `find data/raw/ippl -type d -name...`**

Definition Location

loc_001 - Where is FFT initialization implemented in IPPL?

- Verdict: ****Incorrect****
- Correct reference answer: FFT initialization lives in `src/FFT/FFT.hpp`, especially `FFTBase` construction/setup and the transform-specific constructors that build the heFFTe plans.
- Explanation: The answer does not recover the main Codex reference point or points to insufficiently related implementation details.
- Reference-keyword coverage: `0.17`
- Reference identifier/location hits: `0/3`
- Latency: `33.61s`
- Matched reference signals: `fft, initialization`
- Missing/weak reference signals: `lives, src/fft/fft.hpp, especially, fftbase, construction/setup, transform-specific, constructors, build`
- Claude Code answer excerpt: `To find where FFT (Fast Fourier Transform) initialization is implemented in IPPL, I'll search through the source files for relevant function names and keywords related to FFT initialization. **### Step 1: Search for FFT-related functions and keywords I'll start by searching for "FFT", "fft_init" other related...`**

loc_002 - Where does IPPL set up the MPI environment?

- Verdict: ****Incorrect****
- Correct reference answer: IPPL sets up MPI in `src/Ippl.cpp::initialize()` by constructing `mpi::Environment` and `mpi::Communicator`; the low-level `MPI\_Init` check/call is in `src/Communicate/Environment.cpp`.
- Explanation: The answer does not recover the main Codex reference point or points to insufficiently related implementation details.
- Reference-keyword coverage: `0.20`
- Reference identifier/location hits: `0/9`
- Latency: `7.42s`
- Matched reference signals: `sets, mpi`
- Missing/weak reference signals: `src/ippl.cpp::initialize, constructing, mpi::environment, mpi::communicator, low-level, mpi\_init, check/call, src/communicate/environment.cpp`
- Claude Code answer excerpt: `To determine where IPPL sets up the MPI (Message Passing Interface) environment, I'll explore the relevant files in the `data/raw/ippl` directory. Specifically, I'll look for any initialization or configuration related to MPI. Let's start by searching for mentions of "MPI" within the repository. **### Search for MPI i...`**

loc_003 - Where is Kokkos initialized and finalized inside IPPL?

- Verdict: ****Partial****
- Correct reference answer: Kokkos is initialized and finalized directly in `src/Ippl.cpp` inside `ippl::initialize()` and `ippl::finalize()`.
- Explanation: The answer is related, but it misses the concrete implementation location or symbol emphasized by the Codex reference.
- Reference-keyword coverage: `0.38`
- Reference identifier/location hits: `0/8`
- Latency: `13.29s`
- Matched reference signals: `kokkos, initialized, finalized`
- Missing/weak reference signals: `directly, src/ippl.cpp, inside, ippl::initialize, ippl::finalize, Ippl.cpp, finalize(), initialize()`
- Claude Code answer excerpt: `To determine where Kokkos is initialized and finalized within the IPPL codebase, I'll explore the relevant files. Given that IPPL is not checked into this repository, we will need to rely on any references or examples provided within the existing documentation or configuration files. However, based on common practic...`

loc_004 - Where is the orthogonal recursive bisection load balancer implemented?

- Verdict: ****Partial****
- Correct reference answer: The ORB load balancer is implemented in ``src/Decomposition/OrthogonalRecursiveBisection.hpp`` with declarations in the matching header.
- Explanation: The answer is related, but it misses the concrete implementation location or symbol emphasized by the Codex reference.
- Reference-keyword coverage: ``0.57``
- Reference identifier/location hits: ``0/2``
- Latency: ``39.71s``
- Matched reference signals: ``orb, load, balancer, header``
- Missing/weak reference signals: ``src/decomposition/orthogonalrecursivebisection.hpp, declarations, matching, OrthogonalRecursiveBisection.hpp, src/Decomposition/OrthogonalRecursiveBisection.hpp``
- Claude Code answer excerpt: ``To find where the orthogonal recursive bisection (ORB) load balancer is implemented in the IPPL source tree, I'll explore the relevant files and symbols using the read-only tools. ### Step 1: Search for Relevant Terms I'll start by searching for "orthogonal recursive bisection" or "ORB" in the `data/raw/ippl` direct...``

loc_005 - Where are boundary conditions applied to a Field?

- Verdict: ****Incorrect****
- Correct reference answer: Field BCs are attached in ``Field::setFieldBC()`` and actually applied through ``BConds::apply()`` and the concrete face classes in ``BcTypes.hpp``.
- Explanation: The answer does not recover the main Codex reference point or points to insufficiently related implementation details.
- Reference-keyword coverage: ``0.25``
- Reference identifier/location hits: ``0/7``
- Latency: ``38.51s``
- Matched reference signals: ``field, applied, classes``
- Missing/weak reference signals: ``bcs, attached, field::setfieldbc, actually, through, bconds::apply, concrete, face``
- Claude Code answer excerpt: ``To determine where boundary conditions are applied to a `Field` in the IPPL source tree, I'll explore the relevant files and symbols. Given that this is a scientific computing codebase, we should look for classes or functions related to fields and boundary conditions. ### Step 1: Identify Relevant Files First, let's...``

loc_006 - Where is the CIC (cloud-in-cell) scatter implemented?

- Verdict: ****Partial****
- Correct reference answer: CIC scatter is implemented in ``src/Interpolation/CIC.hpp``, primarily ``scatterToPoint()`` and ``scatterToField()``.
- Explanation: The answer is related, but it misses the concrete implementation location or symbol emphasized by the Codex reference.
- Reference-keyword coverage: ``0.33``
- Reference identifier/location hits: ``0/4``
- Latency: ``26.78s``
- Matched reference signals: ``cic, scatter``
- Missing/weak reference signals: ``src/interpolation/cic.hpp, primarily, scattertopoint, scattertofield, CIC.hpp, scatterToField(), scatterToPoint(), src/Interpolation/CIC.hpp``
- Claude Code answer excerpt: ``To determine where the CIC (cloud-in-cell) scatter is implemented in the IPPL codebase, I'll explore relevant files and symbols under `data/raw/ippl`. ### Step 1: Identify Relevant Files First, I'll look for any mention of "CIC" or "cloud-in-cell" within the source files. #### Search for "CIC" ``bash find data/raw/...``

loc_007 - Where is the CIC gather implemented?

- Verdict: ****Partial****
- Correct reference answer: CIC gather is implemented in ``src/Interpolation/CIC.hpp``, primarily ``gatherFromPoint()`` and ``gatherFromField()``.
- Explanation: The answer is related, but it misses the concrete implementation location or symbol emphasized by the Codex reference.
- Reference-keyword coverage: ``0.33``
- Reference identifier/location hits: ``0/4``
- Latency: ``40.25s``
- Matched reference signals: ``cic, gather``
- Missing/weak reference signals: ``src/interpolation/cic.hpp, primarily, gatherfrompoint, gatherfromfield, CIC.hpp, gatherFromField(), gatherFromPoint(), src/Interpolation/CIC.hpp``
- Claude Code answer excerpt: ``To determine where the **CIC gather** is implemented in the IPPL source code, I'll explore the `data/raw/ippl` directory using the available tools. Since the repository documentation does not specify an exact file or symbol for "CIC gather," I will search for relevant keywords and symbols that might indicate its imp...``

loc_008 - Where is halo exchange for BareField implemented?

- Verdict: ****Partial****

- Correct reference answer: Halo exchange for `BareField` is surfaced in `src/Field/BareField.hpp` and actually performed in `src/Field/HaloCells.hpp`.
- Explanation: The answer overlaps with the Codex reference, but misses important scope, implementation detail, or exact terminology.
- Reference-keyword coverage: `0.38`
- Reference identifier/location hits: `1/5`
- Latency: `18.68s`
- Matched reference signals: `halo, exchange, barefield, BareField`
- Missing/weak reference signals: `surfaced, src/field/barefield.hpp, actually, performed, src/field/halocells.hpp, BareField.hpp, HaloCells.hpp, src/Field/BareField.hpp`
- Claude Code answer excerpt: `To determine where the halo exchange for `BareField` is implemented in the IPPL source code, I'll explore the relevant files and symbols under the `data/raw/ippl` directory. ### Step 1: Search for Relevant Files First, I'll search for files that might contain information about halo exchanges or `BareField`. ```bash...`

loc_009 - Where are the FFT-based open-boundary Poisson solver routines defined?

- Verdict: ****Partial****
- Correct reference answer: The FFT open-boundary Poisson routines are defined in `src/PoissonSolvers/FFTOpenPoissonSolver.h` and `src/PoissonSolvers/FFTOpenPoissonSolver.hpp`.
- Explanation: The answer is related, but it misses the concrete implementation location or symbol emphasized by the Codex reference.
- Reference-keyword coverage: `0.67`
- Reference identifier/location hits: `0/4`
- Latency: `32.09s`
- Matched reference signals: `fft, open-boundary, poisson, defined`
- Missing/weak reference signals: `src/poissonsolvers/fftopenpoissonsolver.h, src/poissonsolvers/fftopenpoissonsolver.hpp, FFTOpenPoissonSolver.h, FFTOpenPoissonSolver.hpp, src/PoissonSolvers/FFTOpenPoissonSolver.h, src/PoissonSolvers/FFTOpenPoissonSolver.hpp`
- Claude Code answer excerpt: `To locate the FFT-based open-boundary Poisson solver routines in the IPPL source code, I'll explore the repository files under `data/raw/ippl` for relevant symbols and definitions. Let's start by searching for terms related to FFT and Poisson solver. ### Searching for FFT and Poisson Solver \*\*Search Terms:\*\* - FFT -...`

loc_010 - Where is the conjugate gradient solver algorithm implemented?

- Verdict: ****Incorrect****
- Correct reference answer: The CG/PCG algorithm implementation is in `src/LinearSolvers/PCG.h`; the Poisson wrapper that instantiates it is `src/PoissonSolvers/PoissonCG.h`.
- Explanation: The answer does not recover the main Codex reference point or points to insufficiently related implementation details.
- Reference-keyword coverage: `0.25`
- Reference identifier/location hits: `0/4`
- Latency: `18.26s`
- Matched reference signals: `algorithm, implementation`
- Missing/weak reference signals: `cg/pcg, src/linearsolvers/pcg.h, poisson, wrapper, instantiates, src/poissonsolvers/poissoncg.h, PCG.h, PoissonCG.h`
- Claude Code answer excerpt: `To find where the conjugate gradient solver algorithm is implemented in the IPPL source tree, I'll explore the relevant files and symbols using the read-only tools provided. Given that this is a numerical method commonly found in linear algebra or solver libraries, we should look for relevant directories like `solve...`

loc_011 - Where is the FEM Poisson solver implemented?

- Verdict: ****Partial****
- Correct reference answer: The FEM Poisson solver is implemented in `src/PoissonSolvers/FEMPoissonSolver.h`.
- Explanation: The answer is related, but it misses the concrete implementation location or symbol emphasized by the Codex reference.
- Reference-keyword coverage: `0.75`
- Reference identifier/location hits: `0/2`
- Latency: `7.52s`
- Matched reference signals: `fem, poisson, solver`
- Missing/weak reference signals: `src/poissonsolvers/fempoissonsolver.h, FEMPoissonSolver.h, src/PoissonSolvers/FEMPoissonSolver.h`
- Claude Code answer excerpt: `To determine where the FEM Poisson solver is implemented in the IPPL (Integrated Plasma Physics Layer) codebase, I will explore the repository files under `data/raw/ippl`. First, I'll search for relevant keywords such as "FEM", "Poisson", and "solver" to locate the implementation. ```search Search: FEM Poisson solve...`

loc_012 - Where is the FDTD Maxwell solver base class defined?

- Verdict: ****Partial****
- Correct reference answer: The FDTD Maxwell solver base class is defined in

`src/MaxwellSolvers/FDTSolverBase.h` with implementation in `FDTSolverBase.hpp`.

- Explanation: The answer is related, but it misses the concrete implementation location or symbol emphasized by the Codex reference.
- Reference-keyword coverage: `0.62`
- Reference identifier/location hits: `0/3`
- Latency: `28.24s`
- Matched reference signals: `fdtd, maxwell, solver, base, defined`
- Missing/weak reference signals: `src/maxwellsolvers/fdtdsolverbase.h, implementation, fdtdsolverbase.hpp, FDTSolverBase.h, FDTSolverBase.hpp, src/MaxwellSolvers/FDTSolverBase.h`
- Claude Code answer excerpt: `To locate the FDTD (Finite-Difference Time-Domain) Maxwell solver base class in the IPPL source tree, I'll search through the relevant directories and files. ### Step 1: Identify Relevant Directories The IPPL codebase typically organizes its components into specific directories based on functionality. For a solver l...`

loc_013 - Where is the LagrangeSpace class defined?

- Verdict: ****Partial****
- Correct reference answer: The `LagrangeSpace` class is defined in `src/FEM/LagrangeSpace.h` with methods implemented in `src/FEM/LagrangeSpace.hpp`.
- Explanation: The answer overlaps with the Codex reference, but misses important scope, implementation detail, or exact terminology.
- Reference-keyword coverage: `0.50`
- Reference identifier/location hits: `1/5`
- Latency: `8.21s`
- Matched reference signals: `lagrangespace, defined, LagrangeSpace`
- Missing/weak reference signals: `src/fem/lagrangespace.h, src/fem/lagrangespace.hpp, LagrangeSpace.h, LagrangeSpace.hpp, src/FEM/LagrangeSpace.h, src/FEM/LagrangeSpace.hpp`
- Claude Code answer excerpt: `To locate the `LagrangeSpace` class definition, I'll explore the source files under `data/raw/ippl`. First, let's search for any occurrences of `LagrangeSpace` in the repository: ``bash grep -r "LagrangeSpace" data/raw/ippl/ `` Let me run this search. file_search: data/raw/ippl/ LagrangeSpace <system-reminder> dat...`

loc_014 - Where is the particle update routine (that moves particles across ranks) implemented?

- Verdict: ****Partial****
- Correct reference answer: Particle migration between ranks is implemented in `ParticleSpatialLayout::update()` in `src/Particle/ParticleSpatialLayout.hpp`.
- Explanation: The answer is related, but it misses the concrete implementation location or symbol emphasized by the Codex reference.
- Reference-keyword coverage: `0.50`
- Reference identifier/location hits: `0/5`
- Latency: `23.13s`
- Matched reference signals: `particle, between, ranks`
- Missing/weak reference signals: `migration, particlespatiallayout::update, src/particle/particlespatiallayout.hpp, ParticleSpatialLayout.hpp, ParticleSpatialLayout::update, ParticleSpatialLayout::update(), src/Particle/ParticleSpatialLayout.hpp, update()`
- Claude Code answer excerpt: `To determine where the particle update routine that moves particles across ranks is implemented in the IPPL source tree, I'll explore the relevant files and symbols within `data/raw/ippl`. ### Step 1: Search for Relevant Files First, let's search for any file names or directories that might contain information about...`

loc_015 - Where are IPPL timers defined?

- Verdict: ****Partial****
- Correct reference answer: IPPL timers are defined by the singleton timing layer in `src/Utility/IpplTimings.h/.cpp`, not just the low-level `Timer` helper.
- Explanation: The answer overlaps with the Codex reference, but misses important scope, implementation detail, or exact terminology.
- Reference-keyword coverage: `0.27`
- Reference identifier/location hits: `1/3`
- Latency: `26.94s`
- Matched reference signals: `timers, defined, timer, Timer`
- Missing/weak reference signals: `singleton, timing, layer, src/utility/ippltimings.h/.cpp, not, just, low-level, helper`
- Claude Code answer excerpt: `To determine where IPPL timers are defined, I'll explore the `data/raw/ippl` directory for relevant files and symbols related to timers. Let's start by searching for timer-related keywords in the source code. ### Step 1: Search for Timer Definitions Let's search for terms like "timer", "Timer", "TIMESTAMP", etc., in...`

Class Responsibility

cls_001 - What is the main responsibility of the BareField class?

- Verdict: ****Correct****
- Correct reference answer: ``BareField`` is the core distributed field storage class: it owns the data view, domain metadata, layout link, and halo operations.
- Explanation: The answer captures the central facts of the Codex reference and includes the important technical nouns, symbols, or implementation locations.
- Reference-keyword coverage: ``0.15``
- Reference identifier/location hits: ``1/1``
- Latency: ``28.98s``
- Matched reference signals: ``barefield, field, BareField``
- Missing/weak reference signals: ``core, distributed, storage, owns, view, domain, metadata, layout``
- Claude Code answer excerpt: ``To determine the main responsibility of the `BareField` class in the IPPL (Integrated Plasma Physics Laboratory) codebase, I will explore the relevant files and symbols under `data/raw/ippl`. First, let's search for mentions of `BareField` to understand where it is defined and used. ### Search for `BareField` Let's...``

cls_002 - What is the main responsibility of the Field class compared to BareField?

- Verdict: ****Correct****
- Correct reference answer: ``Field`` extends ``BareField`` with a mesh, boundary-condition container, and mesh-aware integral/average semantics.
- Explanation: The answer captures the central facts of the Codex reference and includes the important technical nouns, symbols, or implementation locations.
- Reference-keyword coverage: ``0.33``
- Reference identifier/location hits: ``2/2``
- Latency: ``23.59s``
- Matched reference signals: ``field, extends, barefield, BareField, Field``
- Missing/weak reference signals: ``mesh, boundary-condition, container, mesh-aware, integral/average, semantics``
- Claude Code answer excerpt: ``To determine the main responsibilities of the `Field` class compared to the `BareField` class in IPPL, I will explore the source files under `data/raw/ippl`. ### Step 1: Locate Relevant Files I'll start by searching for `Field` and `BareField` within the repository. #### Search for `Field` ```bash Search path=data/r...``

cls_003 - What does the FieldLayout class do and what are its main template parameters?

- Verdict: ****Partial****
- Correct reference answer: ``FieldLayout<Dim>`` distributes a global ``NDIndex<Dim>`` over ranks and stores local domains, neighbors, and communication ranges; its only template parameter is ``Dim``.
- Explanation: The answer overlaps with the Codex reference, but misses important scope, implementation detail, or exact terminology.
- Reference-keyword coverage: ``0.12``
- Reference identifier/location hits: ``1/3``
- Latency: ``37.00s``
- Matched reference signals: ``fieldlayout, template, Dim``
- Missing/weak reference signals: ``dim, distributes, global, ndindex, ranks, stores, local, domains``
- Claude Code answer excerpt: ``To answer the question about the `FieldLayout` class and its main template parameters, I'll explore the relevant files in the IPPL source tree under `data/raw/ippl`. ### Step 1: Locate the `FieldLayout` Class First, let's find where the `FieldLayout` class is defined. ```bash find data/raw/ippl -name '*.h' | xargs g...``

cls_004 - What is the purpose of the ParticleBase class and how is it meant to be used?

- Verdict: ****Correct****
- Correct reference answer: ``ParticleBase`` is the user-extensible base container for particles; derived classes add attributes and register them so layout-driven migration can move them.
- Explanation: The answer captures the central facts of the Codex reference and includes the important technical nouns, symbols, or implementation locations.
- Reference-keyword coverage: ``0.50``
- Reference identifier/location hits: ``1/1``
- Latency: ``22.07s``
- Matched reference signals: ``particlebase, base, particles, derived, classes, them, ParticleBase``
- Missing/weak reference signals: ``user-extensible, container, attributes, register, layout-driven, migration, move``
- Claude Code answer excerpt: ``To answer the question about the `ParticleBase` class, I'll explore the relevant files in the IPPL source tree located under `data/raw/ippl`. Given that this directory isn't checked into the repository, I will assume that it contains the full source code for IPPL. ### Step 1: Identify Relevant Files First, we need t...``

cls_005 - What does the ParticleSpatialLayout class do?

- Verdict: ****Correct****
- Correct reference answer: ``ParticleSpatialLayout`` decides which spatial region/rank owns each particle and performs the migration/update workflow after particles move.
- Explanation: The answer captures the central facts of the Codex reference and includes the important technical

nouns, symbols, or implementation locations.

- Reference-keyword coverage: `0.38`
- Reference identifier/location hits: `1/1`
- Latency: `22.03s`
- Matched reference signals: `particlespatiallayout, spatial, each, particle, particles, ParticleSpatialLayout`
- Missing/weak reference signals: `decides, region/rank, owns, performs, migration/update, workflow, after, move`
- Claude Code answer excerpt: `To determine what the `ParticleSpatialLayout` class does, I'll explore the IPPL source tree located in `data/raw/ippl`. Specifically, I'll look for files containing this class and examine its definition and usage. Let's start by searching for the class name in the repository. ### Searching for `ParticleSpatialLayout...`

cls_006 - What is the responsibility of UniformCartesian?

- Verdict: ****Correct****
- Correct reference answer: `UniformCartesian` represents a uniform Cartesian mesh and exposes spacing/origin/volume and vertex-position geometry queries.
- Explanation: The answer captures the central facts of the Codex reference and includes the important technical nouns, symbols, or implementation locations.
- Reference-keyword coverage: `0.30`
- Reference identifier/location hits: `1/1`
- Latency: `31.18s`
- Matched reference signals: `uniformcartesian, uniform, cartesian, UniformCartesian`
- Missing/weak reference signals: `represents, mesh, exposes, spacing/origin/volume, vertex-position, geometry, queries`
- Claude Code answer excerpt: `To determine the responsibility of `UniformCartesian`, I'll explore the relevant files in the IPPL source tree located at `data/raw/ippl`. First, I'll search for occurrences of `UniformCartesian` to understand its context and role. Let's start by finding where `UniformCartesian` is defined and used. ```bash find dat...`

cls_007 - What does the Mesh base class define?

- Verdict: ****Partial****
- Correct reference answer: `Mesh` is the abstract base interface for meshes: origin, grid size, spacing, vertex positions, and volume queries.
- Explanation: The answer contains some relevant information but also indicates uncertainty or inability to fully answer.
- Reference-keyword coverage: `0.31`
- Reference identifier/location hits: `1/1`
- Latency: `19.67s`
- Matched reference signals: `mesh, abstract, base, interface, Mesh`
- Missing/weak reference signals: `meshes, origin, grid, size, spacing, vertex, positions, volume`
- Claude Code answer excerpt: `To determine what the `Mesh` base class defines in the IPPL (Innovative Parallel Particle Lattice) source code, I'll explore the relevant files within `data/raw/ippl`. First, I'll search for files containing the definition of the `Mesh` class. --- ****Tool:**** Search ```bash find data/raw/ippl -name "*.h" -o -name "*.h...`

cls_008 - What is the responsibility of the Communicator class in ippl::mpi?

- Verdict: ****Incorrect****
- Correct reference answer: `ippl::mpi::Communicator` wraps MPI point-to-point, collectives, tag handling, and managed communication buffers.
- Explanation: The answer does not recover the main Codex reference point or points to insufficiently related implementation details.
- Reference-keyword coverage: `0.20`
- Reference identifier/location hits: `0/1`
- Latency: `21.32s`
- Matched reference signals: `mpi, communication`
- Missing/weak reference signals: `ippl::mpi::communicator, wraps, point-to-point, collectives, tag, handling, managed, buffers`
- Claude Code answer excerpt: `To determine the responsibilities of the `Communicator` class within the `ippl::mpi` namespace, I'll explore the relevant files under `data/raw/ippl`. Given that this class pertains to MPI (Message Passing Interface) communication, it is likely found in files related to parallel computing or message passing. Let's s...`

cls_009 - What is the role of the Environment class in ippl::mpi?

- Verdict: ****Partial****
- Correct reference answer: `ippl::mpi::Environment` manages MPI environment lifetime checks/teardown and provides an abort hook.
- Explanation: The answer overlaps with the Codex reference, but misses important scope, implementation detail, or exact terminology.
- Reference-keyword coverage: `0.25`

- Reference identifier/location hits: `0/1`
- Latency: `20.27s`
- Matched reference signals: `mpi, environment`
- Missing/weak reference signals: `ippl::mpi::environment, manages, lifetime, checks/teardown, abort, hook, ippl::mpi::Environment`
- Claude Code answer excerpt: `To answer the question about the role of the `Environment` class in `ippl::mpi`, I'll explore the relevant files and symbols within the IPPL source tree under `data/raw/ippl`. ### Step 1: Locate Relevant Files I'll start by searching for files that contain mentions of `Environment` and `ippl::mpi`. ``bash fileSearch...`

cls_010 - What does the Archive class do in the Communicate module?

- Verdict: ****Correct****
- Correct reference answer: `Archive` serializes and deserializes Kokkos views and vector-valued views into raw byte buffers for MPI transfer.
- Explanation: The answer captures the central facts of the Codex reference and includes the important technical nouns, symbols, or implementation locations.
- Reference-keyword coverage: `0.08`
- Reference identifier/location hits: `1/1`
- Latency: `278.80s`
- Matched reference signals: `archive, Archive`
- Missing/weak reference signals: `serializes, deserializes, kokkos, views, vector-valued, raw, byte, buffers`
- Claude Code answer excerpt: `To answer the question about the `Archive` class in the `Communicate` module of the IPPL source tree, I'll explore the relevant files under `data/raw/ippl`. ### Step 1: Locate the Communicate Module First, let's identify where the `Communicate` module is located in the IPPL source tree. **\*\*Tool Call:\*\*** Search for the...`

cls_011 - What is the role of the BConds class?

- Verdict: ****Correct****
- Correct reference answer: `BConds` is the container of boundary-condition faces and orchestrates neighbor discovery, BC application, and ghost-to-physical assignment.
- Explanation: The answer captures the central facts of the Codex reference and includes the important technical nouns, symbols, or implementation locations.
- Reference-keyword coverage: `0.10`
- Reference identifier/location hits: `1/1`
- Latency: `28.47s`
- Matched reference signals: `bconds, BConds`
- Missing/weak reference signals: `container, boundary-condition, faces, orchestrates, neighbor, discovery, application, ghost-to-physical`
- Claude Code answer excerpt: `To determine the role of the `BConds` class in the IPPL benchmark, I'll explore the relevant files and symbols within the `data/raw/ippl` directory. ### Step 1: Search for the `BConds` Class Let's find where the `BConds` class is defined and used. ``bash # Find all files containing the BConds class definition or us...`

cls_012 - What is the role of the BaseManager class?

- Verdict: ****Partial****
- Correct reference answer: `BaseManager` provides the simulation loop skeleton: `pre\_run`, `pre\_step`, `advance`, `post\_step`, and `run(nt)`.
- Explanation: The answer overlaps with the Codex reference, but misses important scope, implementation detail, or exact terminology.
- Reference-keyword coverage: `0.33`
- Reference identifier/location hits: `1/6`
- Latency: `21.98s`
- Matched reference signals: `basemanager, simulation, run, BaseManager`
- Missing/weak reference signals: `loop, skeleton, pre\_run, pre\_step, advance, post\_step, run(nt)`
- Claude Code answer excerpt: `To determine the role of the `BaseManager` class in the IPPL benchmark, I'll explore the relevant files within the `data/raw/ippl` directory. ### Step 1: Search for `BaseManager` in the repository Let's search for any mentions of `BaseManager` to see where it is used and defined. ``bash search "BaseManager" data/ra...`

cls_013 - What does the Index class represent and what operations does it support?

- Verdict: ****Partial****
- Correct reference answer: `Index` represents a regular strided integer range/slice and supports range-style arithmetic and comparison operations used in domain descriptions.
- Explanation: The answer contains some relevant information but also indicates uncertainty or inability to fully answer.
- Reference-keyword coverage: `0.31`
- Reference identifier/location hits: `1/1`
- Latency: `32.51s`
- Matched reference signals: `index, represents, supports, operations, Index`

- Missing/weak reference signals: `regular, strided, integer, range/slice, range-style, arithmetic, comparison, domain`
- Claude Code answer excerpt: `To answer the question about the `Index` class, I'll need to explore the source files in `data/raw/ippl`. Given that the IPPL source tree is not checked into the repository, I will assume you have it cloned and accessible at `data/raw/ippl`. Let's search for any mentions of the `Index` class within the IPPL source c...`

Algorithm

alg_001 - How does the FFT-based open-boundary Poisson solver compute the electrostatic potential?

- Verdict: ****Correct****
- Correct reference answer: The FFT open-boundary solver embeds/extends the RHS onto larger domains, FFTs both the density and a precomputed Green's function, multiplies in Fourier space, inverse transforms, and optionally computes gradients/Hessians.
- Explanation: The answer captures the central facts of the Codex reference and includes the important technical nouns, symbols, or implementation locations.
- Reference-keyword coverage: `0.52`
- Latency: `114.89s`
- Matched reference signals: `fft, open-boundary, solver, density, green, multiplies, fourier, space`
- Missing/weak reference signals: `embeds/extends, rhs, onto, larger, domains, ffts, both, precomputed`
- Claude Code answer excerpt: `To answer how the FFT-based open-boundary Poisson solver computes the electrostatic potential in IPPL, I'll need to explore the relevant files and symbols within the `data/raw/ippl` directory. Let's start by identifying any files that might contain the implementation of the FFT-based open-boundary Poisson solver. A...`

alg_002 - How does the FFT-based periodic Poisson solver work in IPPL?

- Verdict: ****Partial****
- Correct reference answer: The periodic FFT solver FFTs the charge density, divides each Fourier mode by $|k|^2$ (with the zero mode handled separately), then inverse transforms; gradients are computed by multiplying by $ik/|k|^2$ before the inverse FFT.
- Explanation: The answer overlaps with the Codex reference, but misses important scope, implementation detail, or exact terminology.
- Reference-keyword coverage: `0.42`
- Reference identifier/location hits: `0/2`
- Latency: `101.55s`
- Matched reference signals: `periodic, fft, solver, density, fourier, then, inverse, transforms`
- Missing/weak reference signals: `ffts, charge, divides, each, mode, zero, handled, separately`
- Claude Code answer excerpt: `To answer how the FFT-based periodic Poisson solver works in IPPL, I'll explore the relevant files and symbols within `data/raw/ippl`. #### Step 1: Locate Relevant Files I'll start by searching for files related to "FFT" and "Poisson" within the IPPL source code. ##### Search Tool: ``bash search -i "fft|poisson" data...`

alg_003 - How does the truncated Green periodic Poisson solver differ from the standard periodic c

- Verdict: ****Incorrect****
- Correct reference answer: The truncated-Green periodic solver replaces the simple $1/|k|^2$ kernel with the FFT of a truncated/erf-smoothed Green's function $\text{forceConstant} * \text{erf}(\alpha r) / r$, then follows the same transform-multiply-inverse pattern.
- Explanation: The answer does not recover the main Codex reference point or points to insufficiently related implementation details.
- Reference-keyword coverage: `0.24`
- Reference identifier/location hits: `0/3`
- Latency: `46.63s`
- Matched reference signals: `periodic, solver, green, then`
- Missing/weak reference signals: `truncated-green, replaces, simple, kernel, fft, truncated/erf-smoothed, forceconstant, erf`
- Claude Code answer excerpt: `To answer this question, I'll need to explore the source files under `data/raw/ippl` and look for relevant symbols and documentation related to the Green periodic Poisson solvers. #### Step 1: Search for Relevant Files First, let's search for files that mention "Green" or "Poisson" i names or content. ``bash...`

alg_004 - How does IPPL's CG solver iterate in the PCG.h implementation?

- Verdict: ****Correct****
- Correct reference answer: In `PCG.h`, the CG iteration forms residual/direction fields, applies the operator, computes `alpha` and `beta` from inner products, updates the solution/residual/direction, and stops on tolerance or iteration count; the preconditioned variant inserts M^{-1} applications on the residual.
- Explanation: The answer captures the central facts of the Codex reference and includes the important technical nouns, symbols, or implementation locations.
- Reference-keyword coverage: `0.43`
- Reference identifier/location hits: `3/4`

- Latency: `169.76s`
- Matched reference signals: `pcg.h, iteration, alpha, beta, products, stops, tolerance, preconditioned`
- Missing/weak reference signals: `forms, residual/direction, fields, applies, operator, computes, inner, updates`
- Claude Code answer excerpt: `To answer how IPPL's Conjugate Gradient (CG) solver iterates in the `PCG.h` implementation, I'll explore the relevant sections of the code. ### Step 1: Locate the PCG Implementation First, I'll search for the `PCG.h` file and any related files that might contain the CG solver implementation. \*\*Search for `PCG.h`:\*\*...`

alg_005 - How is cloud-in-cell interpolation performed in IPPL (both scatter and gather directions)

- Verdict: ****Incorrect****
- Correct reference answer: CIC computes lower/upper weights per axis, scatters with atomic adds to the 2^{Dim} neighboring grid points, and gathers by summing the same weighted stencil back to particles.
- Explanation: The answer does not recover the main Codex reference point or points to insufficiently related implementation details.
- Reference-keyword coverage: `0.26`
- Reference identifier/location hits: `0/1`
- Latency: `30.51s`
- Matched reference signals: `cic, grid, same, back, particles`
- Missing/weak reference signals: `computes, lower/upper, weights, per, axis, scatters, atomic, dim`
- Claude Code answer excerpt: `To answer how cloud-in-cell (CIC) interpolation is performed in IPPL for both scatter and gather operations, we need to explore the source code under `data/raw/ippl`. Since the IPPL source tree is not checked into the repository, I'll outline a structured approach to locate relevant files and symbols. ### Step 1: Id...`

alg_006 - How does the standard FDTD solver advance the electromagnetic field in time?

- Verdict: ****Correct****
- Correct reference answer: The standard FDTD solver advances the source-like field with a second-order finite-difference update using the previous and current time levels plus neighboring cells and source terms, then fills halos and reapplies BCs.
- Explanation: The answer captures the central facts of the Codex reference and includes the important technical nouns, symbols, or implementation locations.
- Reference-keyword coverage: `0.45`
- Latency: `47.90s`
- Matched reference signals: `standard, fdtd, solver, advances, field, finite-difference, update, current`
- Missing/weak reference signals: `source-like, second-order, previous, levels, neighboring, cells, terms, then`
- Claude Code answer excerpt: `To determine how the standard Finite-Difference Time-Domain (FDTD) solver advances the electromagnetic fields in time within the IPPL benchmark, I'll explore relevant files and symbols in the `data/raw/ippl` directory. ### Step 1: Identify Relevant Files Given the category "algorithm" and the topic of FDTD solvers,...`

alg_007 - How does the LagrangeSpace class evaluate degrees of freedom per element?

- Verdict: ****Correct****
- Correct reference answer: `LagrangeSpace` evaluates element DOFs by mapping local element indices to global DOF indices, then evaluates order-1 reference-element shape functions and gradients on those local DOFs.
- Explanation: The answer captures the central facts of the Codex reference and includes the important technical nouns, symbols, or implementation locations.
- Reference-keyword coverage: `0.25`
- Reference identifier/location hits: `1/1`
- Latency: `40.81s`
- Matched reference signals: `lagrangespace, evaluates, element, LagrangeSpace`
- Missing/weak reference signals: `dofs, mapping, local, indices, global, dof, then, order-1`
- Claude Code answer excerpt: `To answer how the `LagrangeSpace` class evaluates degrees of freedom per element in IPPL (Integrated Particle Physics Laboratory), I will explore the relevant files and symbols in the repository. First, let's identify where the `LagrangeSpace` class is defined and used within the codebase. We'll start by searching f...`

alg_008 - How is Gauss-Jacobi quadrature implemented and used in IPPL FEM?

- Verdict: ****Partial****
- Correct reference answer: Gauss-Jacobi quadrature computes nodes/weights by Newton iteration on Jacobi polynomial roots, starting from Chebyshev or LehrFEM-style guesses, and uses the resulting rule on reference elements in FEM.
- Explanation: The answer overlaps with the Codex reference, but misses important scope, implementation detail, or exact terminology.
- Reference-keyword coverage: `0.22`
- Latency: `42.36s`
- Matched reference signals: `gauss-jacobi, quadrature, computes, fem`
- Missing/weak reference signals: `nodes/weights, newton, iteration, jacobi, polynomial, roots, starting, chebyshev`

- Claude Code answer excerpt: `To answer the question about how Gauss-Jacobi quadrature is implemented and used in IPPL FEM, I'll explore the relevant files within the `data/raw/ippl` directory. Given that the codebase might be large, I will start by searching for relevant keywords like "Gauss-Jacobi" and "quadrature". Let's use the search tool t...`

alg_009 - How does the orthogonal recursive bisection decide where to cut the domain?

- Verdict: ****Incorrect****
- Correct reference answer: ORB chooses a cut axis from the domain geometry and then finds a weight-balanced cut/median so recursive splits distribute load more evenly.
- Explanation: The answer does not recover the main Codex reference point or points to insufficiently related implementation details.
- Reference-keyword coverage: `0.19`
- Latency: `29.63s`
- Matched reference signals: `cut, domain, recursive`
- Missing/weak reference signals: `orb, chooses, axis, geometry, then, finds, weight-balanced, cut/median`
- Claude Code answer excerpt: `To answer the question about how the orthogonal recursive bisection decides where to cut the domain in IPPL (Integrated Parallel Processing Library), I'll explore the relevant files and symbols within `data/raw/ippl`. #### Exploration of Relevant Files 1. **\*\*`src/geometry/Partition.h`\*\***: - This file contains definitio...`

alg_010 - How does inverse transform sampling generate particles from a given distribution in IPPL?

- Verdict: ****Partial****
- Correct reference answer: Inverse transform sampling maps uniform random values in CDF space to samples by estimating the inverse and refining it with Newton-Raphson, dimension by dimension.
- Explanation: The answer overlaps with the Codex reference, but misses important scope, implementation detail, or exact terminology.
- Reference-keyword coverage: `0.38`
- Latency: `306.22s`
- Matched reference signals: `inverse, transform, sampling, random, cdf`
- Missing/weak reference signals: `maps, uniform, values, space, samples, estimating, refining, newton-raphson`
- Claude Code answer excerpt: `To answer how inverse transform sampling generates particles from a given distribution in IPPL, I'll explore the relevant files and symbols within the `data/raw/ippl` directory. #### Step 1: Locate Relevant Files First, let's identify files related to inverse transform sampling or particle generation. We can use the...`

alg_011 - How does the FFT class interface with heFFTe to perform complex-to-complex and real-to-complex transforms?

- Verdict: ****Incorrect****
- Correct reference answer: The FFT layer builds heFFTe boxes/plans in `setup()`, copies IPPL field data into the required views/subviews, calls heFFTe forward/backward transforms, and copies results back for CC/RC/DST/DCT variants.
- Explanation: The answer does not recover the main Codex reference point or points to insufficiently related implementation details.
- Reference-keyword coverage: `0.26`
- Reference identifier/location hits: `0/1`
- Latency: `41.70s`
- Matched reference signals: `fft, heffte, calls, transforms`
- Missing/weak reference signals: `layer, builds, boxes/plans, setup, copies, field, required, views/subviews`
- Claude Code answer excerpt: `To answer how the `FFT` class interfaces with heFFTe for performing complex-to-complex and real-to-complex transforms, I will explore the relevant files in the IPPL source tree under `data/raw/ippl`. #### Step 1: Identify Relevant Files First, I'll look for files that mention `FFT`, `heFFTe`, and related transform op...`

alg_012 - How does IPPL's preconditioned CG use the preconditioner defined in Preconditioner.h?

- Verdict: ****Correct****
- Correct reference answer: PCG applies a chosen preconditioner from `Preconditioner.h` to the residual, uses the preconditioned residual in its inner products and direction updates, and iterates until tolerance/max iterations.
- Explanation: The answer captures the central facts of the Codex reference and includes the important technical nouns, symbols, or implementation locations.
- Reference-keyword coverage: `0.38`
- Reference identifier/location hits: `1/1`
- Latency: `34.66s`
- Matched reference signals: `applies, preconditioner, preconditioner.h, residual, preconditioned, Preconditioner.h`
- Missing/weak reference signals: `pcg, chosen, inner, products, direction, updates, iterates, until`
- Claude Code answer excerpt: `To answer how IPPL's preconditioned Conjugate Gradient (CG) uses the preconditioner defined in `Preconditioner.h`, I'll explore the relevant files and symbols within the `data/raw/ippl` directory. #### Key Files and Symbols 1. **\*\*`Preconditioner.h`****: - This file contains the

definition of the `Preconditioner` class....`

Data Flow

flow_001 - How does charge density flow from particles to the grid in a typical PIC step in IPPL?

- Verdict: ****Correct****
- Correct reference answer: In the PIC workflow, particle charges and positions are deposited onto the grid with CIC (`scatter`) or FEM assembly (`assemble\_rhs\_from\_particles`), accumulated into `rho`, and then normalized/background-shifted in the manager before solving.
- Explanation: The answer captures the central facts of the Codex reference and includes the important technical nouns, symbols, or implementation locations.
- Reference-keyword coverage: `0.50`
- Reference identifier/location hits: `0/3`
- Latency: `37.59s`
- Matched reference signals: `pic, particle, charges, positions, deposited, onto, grid, cic`
- Missing/weak reference signals: `workflow, scatter, fem, assemble, assemble\_rhs\_from\_particles, accumulated, rho, then`
- Claude Code answer excerpt: `To answer the question about how charge density flows from particles to the grid in a typical Particle-In-Cell (PIC) step in IPPL, we need to explore the source code under `data/raw/ippl` and relevant documentation. Here's what I found: **### Key Files and Symbols** 1. **\*\*Charge Density Deposition:\*\*** - **\*\*File:\*\*** `data/ra...`

flow_002 - How does the electric field flow from the grid back to particles after a Poisson solve?

- Verdict: ****Partial****
- Correct reference answer: After the field solve, the manager gathers the electric field back to particles with CIC `gather(...)` or FEM `interpolate\_grad\_to\_diracs(...)` , storing particle-local `E` values for the push step.
- Explanation: The answer overlaps with the Codex reference, but misses important scope, implementation detail, or exact terminology.
- Reference-keyword coverage: `0.44`
- Reference identifier/location hits: `0/2`
- Latency: `33.55s`
- Matched reference signals: `after, field, solve, electric, back, particles, step`
- Missing/weak reference signals: `manager, gathers, cic, gather, fem, interpolate\_grad\_to\_diracs, storing, particle-local`
- Claude Code answer excerpt: `To answer how the electric field flows from the grid back to particles after a Poisson solve in IPPL, I'll explore the relevant files and symbols within the `data/raw/ippl` directory. **### Step 1: Identify Key Files** First, we need to identify key files related to the Poisson solver and particle interaction with the e...`

flow_003 - How are halo cells exchanged between neighboring MPI ranks for a BareField?

- Verdict: ****Incorrect****
- Correct reference answer: Halo exchange uses `FieldLayout` neighbor/range metadata to pack field subviews, send/receive them with MPI, unpack into ghost or physical regions, and handle serial-periodic dimensions locally.
- Explanation: The answer does not recover the main Codex reference point or points to insufficiently related implementation details.
- Reference-keyword coverage: `0.21`
- Reference identifier/location hits: `0/1`
- Latency: `42.20s`
- Matched reference signals: `halo, exchange, field, mpi`
- Missing/weak reference signals: `fieldlayout, neighbor/range, metadata, pack, subviews, send/receive, them, unpack`
- Claude Code answer excerpt: `To answer the question about how halo cells are exchanged between neighboring MPI ranks for a `BareField` in IPPL, I will explore the relevant files and symbols in the `data/raw/ippl` directory. **### Step 1: Identify Relevant Files** Given that we're dealing with data flow and MPI communication, files related to parall...`

flow_004 - How are particles migrated between MPI ranks by ParticleSpatialLayout?

- Verdict: ****Partial****
- Correct reference answer: `ParticleSpatialLayout::update()` applies particle BCs, locates destination ranks, advertises receive counts with RMA, sends all registered attributes, destroys invalid local particles, and receives the incoming ones.
- Explanation: The answer overlaps with the Codex reference, but misses important scope, implementation detail, or exact terminology.
- Reference-keyword coverage: `0.36`
- Reference identifier/location hits: `0/3`
- Latency: `35.63s`
- Matched reference signals: `particle, ranks, receive, sends, all, local, particles, receives`

- Missing/weak reference signals: `particlespatiallayout::update, applies, bcs, locates, destination, advertises, counts, rma`
- Claude Code answer excerpt: `To answer how particles are migrated between MPI ranks by `ParticleSpatialLayout`, I'll explore the relevant files in the IPPL source tree located under `data/raw/ippl`.

Step 1: Identify Relevant Files Given that the question pertains to particle migration and MPI, we look for files related to parallelism...

flow_005 - How does FieldLayout decide which index ranges live on which rank?

- Verdict: ****Partial****
- Correct reference answer: `FieldLayout` uses the partitioner and the requested parallel/serial decomposition flags to assign each rank a local `NDIndex`, plus neighbor and communication range tables.
- Explanation: The answer contains some relevant information but also indicates uncertainty or inability to fully answer.
- Reference-keyword coverage: `0.33`
- Reference identifier/location hits: `1/2`
- Latency: `32.56s`
- Matched reference signals: `fieldlayout, decomposition, each, rank, range, FieldLayout`
- Missing/weak reference signals: `partitioner, requested, parallel/serial, flags, assign, local, ndindex, neighbor`
- Claude Code answer excerpt: `To determine how `FieldLayout` decides which index ranges live on which rank in the IPPL benchmark, I'll explore the relevant source files within the `data/raw/ippl` directory. First, let's identify potential candidates for the `FieldLayout` implementation. Typically, such a class would be found in a file related to...`

flow_006 - How is a ParameterList used to configure the FFT and solver classes?

- Verdict: ****Partial****
- Correct reference answer: `ParameterList` is merged into FFT and solver objects, then consumed by `setup()/initialize()/solve()` for communication mode, pencil/reorder flags, transform direction, solver type, tolerances, preconditioner settings, and output type.
- Explanation: The answer overlaps with the Codex reference, but misses important scope, implementation detail, or exact terminology.
- Reference-keyword coverage: `0.35`
- Reference identifier/location hits: `1/5`
- Latency: `40.64s`
- Matched reference signals: `parameterlist, fft, solver, setup, transform, tolerances, settings, ParameterList`
- Missing/weak reference signals: `merged, objects, then, consumed, initialize, solve, communication, mode`
- Claude Code answer excerpt: `To answer the question about how `ParameterList` is used to configure the FFT and solver classes in IPPL, I will explore the relevant files within the `data/raw/ippl` directory. **### Step 1: Identify Relevant Files Given the context of FFT (Fast Fourier Transform) and solver configurations, we should look for files r...**

flow_007 - How does data flow through the BaseManager's pre_step / advance / post_step hooks during simulation?

- Verdict: ****Partial****
- Correct reference answer: `BaseManager::run()` executes `pre\_step()`, `advance()`, and `post\_step()` for each timestep; derived managers use those hooks for deposit/solve/gather, time updates, and output.
- Explanation: The answer overlaps with the Codex reference, but misses important scope, implementation detail, or exact terminology.
- Reference-keyword coverage: `0.25`
- Reference identifier/location hits: `3/6`
- Latency: `31.97s`
- Matched reference signals: `pre\_step, advance, post\_step, hooks, advance(), post\_step(), pre\_step()`
- Missing/weak reference signals: `basemanager::run, executes, each, timestep, derived, managers, use, those`
- Claude Code answer excerpt: `To answer the question about how data flows through the `BaseManager`'s `pre_step`, `advance`, and `post_step` hooks during a simulation, I'll explore the relevant files in the IPPL source tree. **### Key Files to Investigate 1. \*\*`data/raw/ippl/src/core/base_manager.hh`\*\*:** This file likely contains the declaration of...`

flow_008 - How does the FEM solver move data between fields and FEMVector representations?

- Verdict: ****Correct****
- Correct reference answer: FEM solvers move data by assembling field data into FEM load/stiffness representations (`evaluateLoadVector`, `evaluateAx`, `FEMVector`) and then interpolating the solved field quantities back where needed.
- Explanation: The answer captures the central facts of the Codex reference and includes the important technical nouns, symbols, or implementation locations.
- Reference-keyword coverage: `0.50`
- Reference identifier/location hits: `1/3`
- Latency: `31.51s`
- Matched reference signals: `fem, move, field, representations, femvector, back, needed, FEMVector`

- Missing/weak reference signals: `solvers, assembling, load/stiffness, evaluateloadvector, evaluateax, then, interpolating, solved`
- Claude Code answer excerpt: `To answer how the FEM solver moves data between fields and `FEMVector` representations in the IPPL (Integrated Particle Physics Laboratory) codebase, I'll explore relevant files within `data/raw/ippl`. ### Relevant Files: 1. `**`src/fem/FEMField.h`**` – Defines the `FEMField` class. 2. `**`src/fem/FEMVector.h`**` – Defi...`

flow_009 - How does OrthogonalRecursiveBisection redistribute particles after rebalancing?

- Verdict: `**Partial**`
- Correct reference answer: After ORB recomputes the decomposition, particle ownership changes are realized by updating the layout/regions and then migrating particles with the usual particle update path.
- Explanation: The answer overlaps with the Codex reference, but misses important scope, implementation detail, or exact terminology.
- Reference-keyword coverage: `0.29`
- Latency: `34.19s`
- Matched reference signals: `after, particle, then, particles`
- Missing/weak reference signals: `orb, recomputes, decomposition, ownership, changes, realized, updating, layout/regions`
- Claude Code answer excerpt: `To answer the question about how `OrthogonalRecursiveBisection` redistributes particles after rebalancing, I'll explore the relevant files in the IPPL (Integrated Plasma Physics Laboratory) source tree under `data/raw/ippl`. First, let's search for `OrthogonalRecursiveBisection` to find related files and symbols. ##...`

Api Usage

api_001 - How do I create a 3D Field of doubles in IPPL?

- Verdict: `**Partial**`
- Correct reference answer: A typical 3D field is `Field<double, 3, UniformCartesian<double, 3>, Cell>` constructed from a `UniformCartesian` mesh and a `FieldLayout<3>`.
- Explanation: The answer overlaps with the Codex reference, but misses important scope, implementation detail, or exact terminology.
- Reference-keyword coverage: `0.36`
- Reference identifier/location hits: `0/3`
- Latency: `31.52s`
- Matched reference signals: `field, double`
- Missing/weak reference signals: `typical, uniformcartesian, cell, constructed, mesh, fieldlayout, Field<double, 3, UniformCartesian<double, 3>, Cell>, FieldLayout<3>`
- Claude Code answer excerpt: `To answer how to create a 3D Field of doubles in IPPL, I will explore the relevant files and symbols within the `data/raw/ippl` directory. First, I'll search for any mentions of "Field" or related types in the repository: ```bash grep -rni "Field" data/raw/ippl/ ``` ### Step 1: Identify Relevant Files The grep comma...`

api_002 - How do I construct a UniformCartesian mesh with custom spacing and origin?

- Verdict: `**Correct**`
- Correct reference answer: Construct the mesh as `UniformCartesian<double, Dim> mesh(domain, hx, origin);` where `hx` is the spacing vector and `origin` is the mesh origin.`
- Explanation: The answer captures the central facts of the Codex reference and includes the important technical nouns, symbols, or implementation locations.
- Reference-keyword coverage: `0.85`
- Reference identifier/location hits: `1/4`
- Latency: `32.47s`
- Matched reference signals: `construct, mesh, uniformcartesian, double, origin, spacing, vector`
- Missing/weak reference signals: `dim, domain, UniformCartesian<double, Dim> mesh(domain, hx, origin), hx, mesh(domain, hx, origin)`
- Claude Code answer excerpt: `To answer the question about constructing a `UniformCartesian` mesh with custom spacing and origin using IPPL (Innovative Parallel Particle Lattice), we need to explore the relevant source files in the repository. ### Steps to Find Relevant Information 1. `**Identify Relevant Files**`: Look for files related to mesh c...`

api_003 - How do I attach a boundary condition (e.g. periodic) to a Field?

- Verdict: `**Incorrect**`
- Correct reference answer: Create a `BConds<FieldT, Dim>`, assign per-face BC objects such as `PeriodicFace<FieldT>`, then call `field.setFieldBC(bc)`.`
- Explanation: The answer does not recover the main Codex reference point or points to insufficiently related implementation details.
- Reference-keyword coverage: `0.15`
- Reference identifier/location hits: `0/4`
- Latency: `29.44s`
- Matched reference signals: `create, such`

- Missing/weak reference signals: `bconds, fieldt, dim, assign, per-face, objects, periodicface, then`
- Claude Code answer excerpt: `To determine how to attach a boundary condition such as periodic to a `Field` in the IPPL (Inter-Platform Physics Library), I'll explore the relevant files and symbols within the `data/raw/ippl` directory. ### Step-by-step Exploration 1. \*\*Identify Relevant Classes/Structures\*\*: - Look for classes or structures name...`

api_004 - How do I define a custom particle class that adds mass and velocity attributes on top of ParticleBase?

- Verdict: ****Correct****
- Correct reference answer: Define a class derived from `ParticleBase<ParticleSpatialLayout<...>>`, add `mass` and `velocity` attributes as members, and register them with `addAttribute()` in the constructor.
- Explanation: The answer captures the central facts of the Codex reference and includes the important technical nouns, symbols, or implementation locations.
- Reference-keyword coverage: `0.42`
- Reference identifier/location hits: `2/4`
- Latency: `7.47s`
- Matched reference signals: `define, particlebase, mass, velocity, attributes`
- Missing/weak reference signals: `derived, particlespatiallayout, members, register, them, addattribute, constructor, ParticleBase<ParticleSpatialLayout<...>>`
- Claude Code answer excerpt: `To define a custom particle class that adds `mass` and `velocity` attributes on top of `ParticleBase`, you need to explore the IPPL (Integrated Particle Physics Library) source code located in the `data/raw/ippl` directory. First, let's identify where `ParticleBase` is defined. We can search for this class within th...`

api_005 - How do I call ippl::initialize and ippl::finalize in a main program?

- Verdict: ****Correct****
- Correct reference answer: Call the free namespace functions `ippl::initialize(argc, argv);` and `ippl::finalize();` from `main`, as shown in the docs/examples.
- Explanation: The answer captures the central facts of the Codex reference and includes the important technical nouns, symbols, or implementation locations.
- Reference-keyword coverage: `0.60`
- Reference identifier/location hits: `7/7`
- Latency: `21.73s`
- Matched reference signals: `call, ippl::initialize, argc, argv, ippl::finalize, main, finalize(), initialize(argc, argv)`
- Missing/weak reference signals: `free, namespace, shown, docs/examples`
- Claude Code answer excerpt: `To answer the question about how to call `ippl::initialize` and `ippl::finalize` in a main program, let's explore the IPPL source code under `data/raw/ippl`. First, we should look for files that might contain examples or usage patterns of these functions. Common places would be header files defining these functions,...`

api_006 - How do I set up and run a forward FFT on an IPPL Field?

- Verdict: ****Correct****
- Correct reference answer: Set up the relevant layouts/fields/parameters, build an `FFT<CCTransform, ...>` or `FFT<RCTransform, ...>` object, and call `transform(FORWARD, ...)` on the field(s).
- Explanation: The answer captures the central facts of the Codex reference and includes the important technical nouns, symbols, or implementation locations.
- Reference-keyword coverage: `0.62`
- Reference identifier/location hits: `0/4`
- Latency: `29.59s`
- Matched reference signals: `set, relevant, fft, call, transform, forward, field`
- Missing/weak reference signals: `layouts/fields/parameters, build, cctransform, rctransform, object, FFT<CCTransform, ...>, FFT<RCTransform, ...>, field(s)`
- Claude Code answer excerpt: `To answer how to set up and run a forward FFT (Fast Fourier Transform) on an IPPL Field, we need to explore the source code within `data/raw/ippl`. Let's start by identifying relevant files and symbols that might contain information about FFT operations. ### Step 1: Identify Relevant Files We can use the Search tool...`

api_007 - How do I construct an FFTPeriodicPoissonSolver and solve for the potential?

- Verdict: ****Partial****
- Correct reference answer: Construct `FFTPeriodicPoissonSolver(lhs, rhs)` (or default construct then set fields/params), ensure the RHS is attached, and call `solve()` to write the potential/gradient outputs.
- Explanation: The answer overlaps with the Codex reference, but misses important scope, implementation detail, or exact terminology.
- Reference-keyword coverage: `0.29`
- Reference identifier/location hits: `0/2`
- Latency: `34.77s`
- Matched reference signals: `construct, fftperiodicpoissonsolver, call, solve`
- Missing/weak reference signals: `lhs, rhs, default, then, set, fields/params, ensure, attached`

- Claude Code answer excerpt: `To answer how to construct an `FFTPeriodicPoissonSolver` and solve for the potential in the IPPL (Incompressible Plasma Physics Laboratory) codebase, we need to explore the relevant files and symbols. ### Step 1: Locate Relevant Classes and Methods First, we will search for classes related to `FFTPeriodicPoissonSolv...`

api_008 - How do I invoke the conjugate gradient Poisson solver on a Field?

- Verdict: ****Incorrect****
- Correct reference answer: Use `PoissonCG<FieldT>` (or the Alpine `FieldSolver` wrapper around it), set its fields/parameters, and call `solve()` so the internal CG/PCG object iterates on the field.
- Explanation: The answer does not recover the main Codex reference point or points to insufficiently related implementation details.
- Reference-keyword coverage: `0.19`
- Reference identifier/location hits: `0/3`
- Latency: `37.90s`
- Matched reference signals: `call, solve, field`
- Missing/weak reference signals: `use, poissoncg, fieldt, alpine, fieldsolver, wrapper, around, set`
- Claude Code answer excerpt: `To answer how to invoke the conjugate gradient Poisson solver on a `Field` in IPPL, let's explore the source code under `data/raw/ippl`. First, we'll look for any mention of the conjugate gradient Poisson solver and see if it is associated with the `Field` class. ### Step 1: Search for Conjugate Gradient Poisson Sol...`

api_009 - How do I sample particles from a normal distribution using IPPL Random utilities?

- Verdict: ****Incorrect****
- Correct reference answer: For Gaussian draws, IPPL exposes both `randn` convenience generation and the more general `InverseTransformSampling` + `NormalDistribution` path.
- Explanation: The answer does not recover the main Codex reference point or points to insufficiently related implementation details.
- Reference-keyword coverage: `0.08`
- Reference identifier/location hits: `0/3`
- Latency: `32.80s`
- Matched reference signals: `generation`
- Missing/weak reference signals: `gaussian, draws, exposes, both, randn, convenience, more, general`
- Claude Code answer excerpt: `To answer how to sample particles from a normal distribution using IPPL Random utilities, let's explore the relevant parts of the IPPL codebase located in `data/raw/ippl`. ### Step 1: Identify Relevant Headers and Source Files First, we need to find the headers and source files that deal with random number generatio...`

api_010 - How do I register particle attributes with ParticleBase so that they are communicated on migration?

- Verdict: ****Correct****
- Correct reference answer: Register each particle attribute with `addAttribute(...)`; migration then serializes and transfers all registered attributes automatically.
- Explanation: The answer captures the central facts of the Codex reference and includes the important technical nouns, symbols, or implementation locations.
- Reference-keyword coverage: `0.62`
- Reference identifier/location hits: `0/1`
- Latency: `32.74s`
- Matched reference signals: `register, particle, attribute, addattribute, migration, all, registered, attributes`
- Missing/weak reference signals: `each, then, serializes, transfers, automatically, addAttribute(...)`
- Claude Code answer excerpt: `To answer how to register particle attributes with `ParticleBase` so that they are communicated on migration in the IPPL (Integrated Plasma Physics Laboratory) codebase, let's explore the relevant files and symbols. ### Step 1: Identify Relevant Files First, we need to find where `ParticleBase` is defined and how pa...`

api_011 - How do I access the MPI rank and size through the IPPL Communicator?

- Verdict: ****Correct****
- Correct reference answer: Access rank and size with `ippl::Comm->rank()` and `ippl::Comm->size()` (or the same methods on a `Communicator` object).
- Explanation: The answer captures the central facts of the Codex reference and includes the important technical nouns, symbols, or implementation locations.
- Reference-keyword coverage: `0.60`
- Reference identifier/location hits: `3/6`
- Latency: `22.01s`
- Matched reference signals: `access, rank, size, communicator, Communicator, rank(), size()`
- Missing/weak reference signals: `ippl::comm-, same, object, ippl::Comm, ippl::Comm->rank(), ippl::Comm->size()`
- Claude Code answer excerpt: `To determine how to access the MPI rank and size through the IPPL Communicator, I'll explore the source files under `data/raw/ippl` for relevant classes and methods. ### Step 1: Identify

Relevant Files Let's search for files related to communicators in the IPPL codebase. ****Tool Call:**** Search for communicator-relate...`

api_012 - How do I use IpplTimings to measure a region of code?

- Verdict: ****Partial****
- Correct reference answer: Get a timer handle with ``IpplTimings::getTimer("name")``, surround the code with ``startTimer()`` / ``stopTimer()``, and print with ``IpplTimings::print()``.
- Explanation: The answer overlaps with the Codex reference, but misses important scope, implementation detail, or exact terminology.
- Reference-keyword coverage: ``0.10``
- Reference identifier/location hits: ``1/8``
- Latency: ``45.09s``
- Matched reference signals: ``print, IpplTimings::print``
- Missing/weak reference signals: ``get, timer, handle, ippltimings::gettimer, name, surround, starttimer, stoptimer``
- Claude Code answer excerpt: ``To understand how to use `IpplTimings` to measure a region of code in IPPL (Integrated Parallel Physics Layer), we need to explore the relevant source files and documentation within the repository. #### Step 1: Identify Relevant Files First, let's search for any mentions of `IpplTimings` in the IPPL source tree under...``

Parallelism And Kokkos

par_001 - How does IPPL achieve performance portability across CPUs and GPUs?

- Verdict: ****Correct****
- Correct reference answer: IPPL gets performance portability by combining Kokkos execution/memory abstractions with MPI and heffTe, so the same code paths can target CPUs and GPUs.
- Explanation: The answer captures the central facts of the Codex reference and includes the important technical nouns, symbols, or implementation locations.
- Reference-keyword coverage: ``0.50``
- Latency: ``41.38s``
- Matched reference signals: ``performance, portability, kokkos, abstractions, target, cpus, gpus``
- Missing/weak reference signals: ``gets, combining, execution/memory, mpi, heffte, same, paths``
- Claude Code answer excerpt: ``To understand how IPPL (Integrated Plasma Physics Library) achieves performance portability across CPUs and GPUs, I'll explore the relevant files in the `data/raw/ippl` directory. #### Key Concepts 1. **Kokkos**: IPPL uses Kokkos as a performance portable programming model to target different architectures like CPUs...``

par_002 - Why does IPPL use Kokkos::View for field storage?

- Verdict: ****Correct****
- Correct reference answer: IPPL uses ``Kokkos::View`` because fields need execution-space-aware storage, subviews, mirrors, and portable kernels across host/device memory spaces.
- Explanation: The answer captures the central facts of the Codex reference and includes the important technical nouns, symbols, or implementation locations.
- Reference-keyword coverage: ``0.57``
- Reference identifier/location hits: ``1/1``
- Latency: ``31.51s``
- Matched reference signals: ``kokkos::view, because, fields, need, storage, across, memory, spaces``
- Missing/weak reference signals: ``execution-space-aware, subviews, mirrors, portable, kernels, host/device``
- Claude Code answer excerpt: ``To understand why IPPL uses `Kokkos::View` for field storage, we need to explore the relevant parts of the IPPL source code and documentation located in `data/raw/ippl`. #### Exploration of Relevant Files 1. **Field Storage Implementation** - The primary class responsible for handling fields is likely defined within...``

par_003 - How does IPPL parallelize loops over field indices using Kokkos RangePolicy?

- Verdict: ****Partial****
- Correct reference answer: Loops are parallelized through ``ippl::getRangePolicy`` / ``createRangePolicy`` and the ``ippl::parallel_for`` wrapper over Kokkos ``RangePolicy`` / ``MDRangePolicy``.
- Explanation: The answer overlaps with the Codex reference, but misses important scope, implementation detail, or exact terminology.
- Reference-keyword coverage: ``0.30``
- Reference identifier/location hits: ``1/5``
- Latency: ``31.04s``
- Matched reference signals: ``loops, kokkos, rangepolicy, RangePolicy``
- Missing/weak reference signals: ``parallelized, through, ippl::getrangepolicy, createrangepolicy, ippl::parallel_for, wrapper, mdrangepolicy, MDRangePolicy``
- Claude Code answer excerpt: ``To answer how IPPL parallelizes loops over field indices using Kokkos `RangePolicy`, I'll explore the relevant parts of the IPPL source tree located in `data/raw/ippl`. Specifically, I will look for any mentions of `Kokkos::RangePolicy` and related constructs. First, let's search for occurrences of `Kokkos::RangePol...``

par_004 - How does IPPL support mixed precision (float vs double) for fields?

- Verdict: ****Partial****

- Correct reference answer: Precision is chosen by the field value type (``Field<float,...>`` vs ``Field<double,...>``), and the codebase also contains mixed-precision examples where fields/particles deliberately use different precisions.

- Explanation: The answer overlaps with the Codex reference, but misses important scope, implementation detail, or exact terminology.

- Reference-keyword coverage: ``0.50``

- Reference identifier/location hits: ``0/2``

- Latency: ``41.77s``

- Matched reference signals: ``precision, field, value, type, float, double, different``

- Missing/weak reference signals: ``chosen, codebase, contains, mixed-precision, examples, fields/particles, deliberately, use``

- Claude Code answer excerpt: ``To determine how IPPL supports mixed precision (float vs. double) for fields, I'll explore the relevant source files and documentation within the `data/raw/ippl` directory. ### Relevant Files 1. **`Field.h`** 2. **`Field.tpp`** 3. **`GridManager.h`** Let's start by examining these files to find mentions of mixed pre...``

par_005 - How does IPPL handle multiple execution spaces like OpenMP and CUDA in the same build?

- Verdict: ****Partial****

- Correct reference answer: Multiple execution spaces are enabled at build time through ``IPPL_PLATFORMS`` and Kokkos/HeFFTe backend flags; IPPL also has container helpers for multi-space data.

- Explanation: The answer overlaps with the Codex reference, but misses important scope, implementation detail, or exact terminology.

- Reference-keyword coverage: ``0.47``

- Reference identifier/location hits: ``0/1``

- Latency: ``22.52s``

- Matched reference signals: ``multiple, execution, spaces, enabled, build, through, flags``

- Missing/weak reference signals: ``time, ippl_platforms, kokkos/heffte, backend, has, container, helpers, multi-space``

- Claude Code answer excerpt: ``To determine how IPPL handles multiple execution spaces like OpenMP and CUDA in the same build, I'll explore relevant files within `data/raw/ippl`. ### Relevant Files: - `include/IPPL/ExecutionSpace.hh` - `src/ExecutionSpace/CMakeLists.txt` #### 1. `include/IPPL/ExecutionSpace.hh` This header file defines the execut...``

par_006 - What is the purpose of ParallelDispatch.h and what helpers does it expose?

- Verdict: ****Partial****

- Correct reference answer: ``ParallelDispatch.h`` wraps Kokkos policies and functors, exposing ``RangePolicy``, ``getRangePolicy``, ``createRangePolicy``, ``parallel_for``, and ``parallel_reduce``.

- Explanation: The answer overlaps with the Codex reference, but misses important scope, implementation detail, or exact terminology.

- Reference-keyword coverage: ``0.27``

- Reference identifier/location hits: ``2/6``

- Latency: ``26.36s``

- Matched reference signals: ``paralleldispatch.h, kokkos, policies, ParallelDispatch.h, parallel_for``

- Missing/weak reference signals: ``wraps, functors, exposing, rangepolicy, getrangepolicy, createrangepolicy, parallel_for, parallel_reduce``

- Claude Code answer excerpt: ``To answer the question about `ParallelDispatch.h` in the IPPL (Innovative Parallel Particle Library) codebase, I will explore the relevant files using the repository tools. ### Step 1: Locate `ParallelDispatch.h` First, I'll search for `ParallelDispatch.h` within the `data/raw/ippl` directory to understand its conte...``

par_007 - How does the Archive class serialize/deserialize Kokkos views for MPI transfer?

- Verdict: ****Correct****

- Correct reference answer: ``Archive`` serializes Kokkos views into one-dimensional byte buffers with Kokkos kernels and ``memcpy``, then deserializes them back into views on receive.

- Explanation: The answer captures the central facts of the Codex reference and includes the important technical nouns, symbols, or implementation locations.

- Reference-keyword coverage: ``0.50``

- Reference identifier/location hits: ``1/2``

- Latency: ``41.17s``

- Matched reference signals: ``archive, serializes, kokkos, views, deserializes, receive, Archive``

- Missing/weak reference signals: ``one-dimensional, byte, buffers, kernels, memcpy, then, them, back``

- Claude Code answer excerpt: ``To answer the question about how the `Archive` class serializes and deserializes Kokkos views for MPI transfer, I'll explore the relevant files in the IPPL repository. ### Step 1: Identify Relevant Files The IPPL codebase contains classes related to serialization and parallelism. Since we are interested in `Archive`...``

par_008 - How are MPI tags managed in IPPL to avoid collisions across subsystems?

- Verdict: ****Correct****
- Correct reference answer: MPI tags are managed with fixed base tags in `Tags.h` and the cyclic tag allocator in `TagMaker`, which `Communicator` inherits.
- Explanation: The answer captures the central facts of the Codex reference and includes the important technical nouns, symbols, or implementation locations.
- Reference-keyword coverage: `0.46`
- Reference identifier/location hits: `2/3`
- Latency: `52.70s`
- Matched reference signals: `mpi, tags, managed, base, tag, Communicator, Tags.h`
- Missing/weak reference signals: `fixed, tags.h, cyclic, allocator, tagmaker, communicator, inherits, TagMaker`
- Claude Code answer excerpt: `To determine how MPI tags are managed in IPPL to avoid collisions across subsystems, I'll explore the relevant files and symbols within the `data/raw/ippl` directory. First, let's search for mentions of MPI tags and any related management strategies. ### Step 1: Search for MPI Tags Let's use the `Search` tool to fin...`

par_009 - What does Kokkos::fence do, and when does IPPL call the wrapper ippl::fence?

- Verdict: ****Partial****
- Correct reference answer: `Kokkos::fence()` waits for outstanding kernels; IPPL exposes `ippl::fence()` as a runtime wrapper in `Ippl.cpp` and also fences internally after many kernel/data-movement steps.
- Explanation: The answer overlaps with the Codex reference, but misses important scope, implementation detail, or exact terminology.
- Reference-keyword coverage: `0.20`
- Reference identifier/location hits: `4/6`
- Latency: `172.03s`
- Matched reference signals: `kokkos::fence, ippl::fence, wrapper, Kokkos::fence, Kokkos::fence(), fence()`
- Missing/weak reference signals: `waits, outstanding, kernels, exposes, runtime, ippl.cpp, fences, internally`
- Claude Code answer excerpt: `To answer the question about `Kokkos::fence` and its usage in IPPL, we need to explore the source files under `data/raw/ippl`. Specifically, we will look for any mentions of `Kokkos::fence` and how they are used within IPPL. ### Step 1: Search for `Kokkos::fence` Let's start by searching for occurrences of `Kokkos::...`

Boundary And Halo

bc_001 - What boundary condition types are provided in BcTypes.h?

- Verdict: ****Correct****
- Correct reference answer: `BcTypes.h` provides periodic, constant, zero, extrapolating, and no-BC face classes.
- Explanation: The answer captures the central facts of the Codex reference and includes the important technical nouns, symbols, or implementation locations.
- Reference-keyword coverage: `0.25`
- Reference identifier/location hits: `1/1`
- Latency: `20.50s`
- Matched reference signals: `bctypes.h, periodic, BcTypes.h`
- Missing/weak reference signals: `constant, zero, extrapolating, no-bc, face, classes`
- Claude Code answer excerpt: `To answer the question about what boundary condition types are provided in `BcTypes.h`, I'll explore the file using the read-only tools available. Let's start by reading the `BcTypes.h` file from the `data/raw/ippl` directory. Read: data/raw/ippl/include/BcTypes.h ### Content of BcTypes.h ```cpp #ifndef __BC_TYPES_H...`

bc_002 - How does a periodic boundary condition differ from a Neumann boundary condition in IPPL?

- Verdict: ****Partial****
- Correct reference answer: Periodic BCs wrap values from the opposite side of the domain; there is no explicit `NeumannFace` class in this codebase, so the closest non-periodic behavior is constant/extrapolating face treatment.
- Explanation: The answer overlaps with the Codex reference, but misses important scope, implementation detail, or exact terminology.
- Reference-keyword coverage: `0.35`
- Reference identifier/location hits: `0/1`
- Latency: `36.93s`
- Matched reference signals: `periodic, wrap, values, opposite, domain, there`
- Missing/weak reference signals: `bcs, side, explicit, neumannface, codebase, closest, non-periodic, behavior`
- Claude Code answer excerpt: `To answer the question about how a periodic boundary condition differs from a Neumann boundary condition in IPPL Fields, I'll explore the relevant files and symbols in the IPPL source tree located at `data/raw/ippl`. ### Step 1: Identify Relevant Files First, let's identify files related to boundary conditions in IP...`

bc_003 - How are ghost cells populated during a halo exchange?

- Verdict: ****Partial****
- Correct reference answer: Ghost cells are populated mainly by `fillHalo()` / `PeriodicFace::apply()`: source slabs are packed, exchanged, and unpacked into ghost regions; `accumulateHalo_noghost()` is a different operation.
- Explanation: The answer overlaps with the Codex reference, but misses important scope, implementation detail, or exact terminology.
- Reference-keyword coverage: ``0.25``
- Reference identifier/location hits: ``0/5``
- Latency: ``30.91s``
- Matched reference signals: ``ghost, cells, populated``
- Missing/weak reference signals: ``mainly, fillhalo, periodicface::apply, source, slabs, packed, exchanged, unpacked``
- Claude Code answer excerpt: ``To answer how ghost cells are populated during a halo exchange in the IPPL (Innovative Parallel Particle Library) benchmark, I'll explore the relevant files and symbols within the `data/raw/ippl` directory. ### Key Files to Investigate 1. **Halo Exchange Implementation:** - `src/mpi/HaloExchange.cpp` - `src/mpi/Halo...``

bc_004 - How are physical-cell values copied into ghost cells for periodic boundary conditions?

- Verdict: ****Partial****
- Correct reference answer: Periodic ghost cells are filled from opposite-side physical cells in `applyPeriodicSerialDim()` and the multi-rank `PeriodicFace::apply()` path.
- Explanation: The answer overlaps with the Codex reference, but misses important scope, implementation detail, or exact terminology.
- Reference-keyword coverage: ``0.36``
- Reference identifier/location hits: ``0/4``
- Latency: ``31.24s``
- Matched reference signals: ``periodic, ghost, cells``
- Missing/weak reference signals: ``filled, opposite-side, physical, applyperiodicserialdim, multi-rank, periodicface::apply, path, PeriodicFace::apply``
- Claude Code answer excerpt: ``To answer how physical-cell values are copied into ghost cells for periodic boundary conditions in the IPPL (Integrated Plasma Physics Laboratory) codebase, I'll explore the relevant files and symbols. ### Step 1: Identify Relevant Files The handling of periodic boundary conditions is likely to be found in files rel...``

bc_005 - How does the absorbing boundary condition in the MaxwellSolvers module damp outgoing waves?

- Verdict: ****Partial****
- Correct reference answer: The absorbing Maxwell BC uses second-order Mur-style formulas with precomputed weights and previous/current/next boundary values plus tangential neighbors to damp outgoing waves.
- Explanation: The answer overlaps with the Codex reference, but misses important scope, implementation detail, or exact terminology.
- Reference-keyword coverage: ``0.33``
- Latency: ``44.94s``
- Matched reference signals: ``absorbing, boundary, values, outgoing, waves``
- Missing/weak reference signals: ``maxwell, second-order, mur-style, formulas, precomputed, weights, previous/current/next, tangential``
- Claude Code answer excerpt: ``To answer the question about how the absorbing boundary condition (ABC) in the `MaxwellSolvers` module damps outgoing waves, I will explore the relevant files and symbols within the IPPL source code located at `data/raw/ippl`. ### Step 1: Identify Relevant Files Given that we are looking for information related to t...``

Numerical Meaning

num_001 - What does the FFT-based Poisson solver physically compute in terms of Coulomb's law?

- Verdict: ****Correct****
- Correct reference answer: The FFT Poisson solvers compute electrostatic potential (and optionally field/gradient) from charge density, i.e. the discretized Coulomb potential corresponding to $-\Delta\phi = \rho$.
- Explanation: The answer captures the central facts of the Codex reference and includes the important technical nouns, symbols, or implementation locations.
- Reference-keyword coverage: ``0.67``
- Reference identifier/location hits: ``0/1``
- Latency: ``33.44s``
- Matched reference signals: ``fft, poisson, solvers, compute, electrostatic, potential, charge, density``
- Missing/weak reference signals: ``optionally, field/gradient, i.e, discretized, corresponding,  $-\Delta\phi = \rho$ ``
- Claude Code answer excerpt: ``To answer the question about what the FFT-based Poisson solver computes in terms of Coulomb's law, I'll explore the relevant files and symbols within the `data/raw/ippl` directory. ### Step 1: Identify Relevant Files I'll search for mentions of "FFT", "Poisson", and related concepts like "Coulomb" to locate the rele...``

num_002 - What is the physical meaning of the fields advanced by the FDTD solver?

- Verdict: ****Correct****
- Correct reference answer: The Maxwell solvers evolve electromagnetic field data: electric and magnetic fields ('E', 'B') driven by a source/four-current field.
- Explanation: The answer captures the central facts of the Codex reference and includes the important technical nouns, symbols, or implementation locations.
- Reference-keyword coverage: `0.64`
- Latency: `31.95s`
- Matched reference signals: `maxwell, electromagnetic, field, electric, magnetic, fields`
- Missing/weak reference signals: `solvers, evolve, driven, source/four-current`
- Claude Code answer excerpt: `To answer the question about the physical meaning of the fields advanced by the FDTD (Finite-Difference Time-Domain) solver in IPPL, I'll explore the relevant parts of the codebase. The FDTD method is typically used to solve Maxwell's equations numerically, and it involves advancing electric and magnetic fields over...`

num_007 - What are the numerical tradeoffs between the FFT and CG Poisson solvers in IPPL?

- Verdict: ****Correct****
- Correct reference answer: FFT Poisson is fast and direct on structured uniform grids with compatible BCs but less flexible; CG/PCG is iterative and may be slower/convergence-limited, but it is more flexible for alternative discretizations, BCs, and preconditioning.
- Explanation: The answer captures the central facts of the Codex reference and includes the important technical nouns, symbols, or implementation locations.
- Reference-keyword coverage: `0.39`
- Latency: `32.28s`
- Matched reference signals: `fft, poisson, fast, but, flexible, iterative, more`
- Missing/weak reference signals: `direct, structured, uniform, grids, compatible, bcs, less, cg/pcg`
- Claude Code answer excerpt: `To address the question about the numerical tradeoffs between the FFT (Fast Fourier Transform) and CG (Conjugate Gradient) Poisson solvers in IPPL, I'll explore the relevant source files and documentation within the `data/raw/ippl` directory. ### Step 1: Identify Relevant Files First, let's search for files that men...`

Build And Install

build_001 - Which third-party libraries does IPPL depend on, and what are they used for?

- Verdict: ****Partial****
- Correct reference answer: Core third-party dependencies are MPI for distributed communication, Kokkos for performance portability, heFFTe for FFT backends/plans, FFTW/CuFFT as optional FFT backends, and GoogleTest for tests.
- Explanation: The answer contains some relevant information but also indicates uncertainty or inability to fully answer.
- Reference-keyword coverage: `0.22`
- Latency: `50.23s`
- Matched reference signals: `third-party, dependencies, mpi, tests`
- Missing/weak reference signals: `core, distributed, communication, kokkos, performance, portability, heffte, fft`
- Claude Code answer excerpt: `To determine which third-party libraries IPPL depends on and their purposes, I'll explore the `data/raw/ippl` directory for relevant configuration files such as `CMakeLists.txt`, `requirements.txt`, or any other build scripts that specify dependencies. Let's start by reading the `CMakeLists.txt` file to identify ext...`

build_002 - What C++ standard does IPPL require, and why?

- Verdict: ****Partial****
- Correct reference answer: IPPL requires C++20, as stated in the README badge and the installation examples/configuration flags.
- Explanation: The answer overlaps with the Codex reference, but misses important scope, implementation detail, or exact terminology.
- Reference-keyword coverage: `0.25`
- Latency: `26.05s`
- Matched reference signals: `requires, c++20`
- Missing/weak reference signals: `stated, readme, badge, installation, examples/configuration, flags`
- Claude Code answer excerpt: `To determine the C++ standard required by IPPL (Inter-Particle Potential Library), I'll explore the repository files under `data/raw/ippl`. ### Step 1: Search for mentions of C++ standards in Makefiles and CMakeLists.txt I'll search for keywords related to C++ standards, such as `CXX_STANDARD`, `CXXFLAGS`, or specif...`

build_003 - How is IPPL configured with CMake to enable a GPU build?

- Verdict: ****Incorrect****
- Correct reference answer: GPU builds are enabled through CMake options like `IPPL\_PLATFORMS=CUDA` or `IPPL\_PLATFORMS="HIP;OPENMP"` plus matching Kokkos/architecture/backend flags (`Kokkos\_ARCH\_\*`, `CMAKE\_CUDA\_ARCHITECTURES`, `Kokkos\_ENABLE\_HIP`, `Heffte\_ENABLE\_ROCM`, etc.).

- Explanation: The answer does not recover the main Codex reference point or points to insufficiently related implementation details.
- Reference-keyword coverage: `0.30`
- Reference identifier/location hits: `0/6`
- Latency: `34.59s`
- Matched reference signals: `gpu, enabled, through, cmake, cuda, flags`
- Missing/weak reference signals: `builds, options, like, ippl\_platforms, hip, openmp, matching, kokkos/architecture/backend`
- Claude Code answer excerpt: `To determine how IPPL (Interactive Particle-in-Cell) is configured with CMake for a GPU build, I will explore the relevant files in the `data/raw/ippl` directory. ### Step 1: Locate CMake Configuration Files First, let's find the main CMake configuration file and any related GPU-specific configurations. **Tool Call:...

build_004 - What options does INSTALLATION.md describe for configuring Kokkos for IPPL?

- Verdict: ****Partial****
- Correct reference answer: `INSTALLATION.md` describes Kokkos version selection plus backend/architecture options used with IPPL builds, including CUDA and HIP/OpenMP examples and `IPPL\_PLATFORMS` combinations.
- Explanation: The answer overlaps with the Codex reference, but misses important scope, implementation detail, or exact terminology.
- Reference-keyword coverage: `0.23`
- Reference identifier/location hits: `1/2`
- Latency: `29.70s`
- Matched reference signals: `installation.md, kokkos, options, INSTALLATION.md`
- Missing/weak reference signals: `describes, version, selection, backend/architecture, builds, cuda, hip/openmp, examples`
- Claude Code answer excerpt: `To answer the question about the options described in `INSTALLATION.md` for configuring Kokkos for IPPL, I will explore the relevant files and documentation within the repository. Let's start by searching for any mentions of `Kokkos` in the `INSTALLATION.md` file. ### Step 1: Search for "Kokkos" in `INSTALLATION.md`...`

build_005 - How do I install IPPL on the EULER cluster according to the documentation?

- Verdict: ****Incorrect****
- Correct reference answer: The EULER install guide says to connect with VPN/SSH, switch to the new software stack, purge modules, load GCC/CMake/CUDA/OpenMPI, clone the repo, configure a build, compile, and test on an interactive node.
- Explanation: The answer does not recover the main Codex reference point or points to insufficiently related implementation details.
- Reference-keyword coverage: `0.18`
- Latency: `24.57s`
- Matched reference signals: `euler, install, guide, build`
- Missing/weak reference signals: `says, connect, vpn/ssh, switch, new, software, stack, purge`
- Claude Code answer excerpt: `To install IPPL (Improved Particle-in-Cell) on the Euler cluster according to the available documentation in the repository, you need to refer to the files and instructions provided within `data/raw/ippl`. However, based on the current repository status, the specific installation guide for the Euler cluster is not d...`

Testing And Workflow

test_002 - What naming conventions does WORKFLOW.md require for variables, constants, and member v

- Verdict: ****Partial****
- Correct reference answer: `WORKFLOW.md` requires camelCase variables, CAPITAL_CASE compile-time constants, and member names suffixed with `\_m`.
- Explanation: The answer overlaps with the Codex reference, but misses important scope, implementation detail, or exact terminology.
- Reference-keyword coverage: `0.10`
- Reference identifier/location hits: `1/2`
- Latency: `8.60s`
- Matched reference signals: `workflow.md, WORKFLOW.md`
- Missing/weak reference signals: `requires, camelcase, variables, capital\_case, compile-time, constants, member, names`
- Claude Code answer excerpt: `To answer your question, I'll explore the `WORKFLOW.md` file within the IPPL source tree located at `data/raw/ippl`. If this file doesn't exist or doesn't contain the required information, I will indicate that clearly. Let's start by searching for `WORKFLOW.md` in the repository. ``` find data/raw/ippl -name "WORKFL...`

test_003 - When should code use Kokkos::sqrt versus std::sqrt according to WORKFLOW.md?

- Verdict: ****Correct****
- Correct reference answer: Use `std::sqrt` in host-only code, and `Kokkos::sqrt` in device-capable code such as `KOKKOS\_INLINE\_FUNCTION` kernels.

- Explanation: The answer captures the central facts of the Codex reference and includes the important technical nouns, symbols, or implementation locations.
- Reference-keyword coverage: `0.38`
- Reference identifier/location hits: `2/3`
- Latency: `32.56s`
- Matched reference signals: `use, std::sqrt, kokkos::sqrt, Kokkos::sqrt`
- Missing/weak reference signals: `host-only, device-capable, such, kokkos\_inline\_function, kernels, KOKKOS\_INLINE\_FUNCTION`
- Claude Code answer excerpt: `To answer the question regarding when code should use `Kokkos::sqrt` versus `std::sqrt` according to `WORKFLOW.md`, I will search for relevant content in that file. Let's start by reading `docs/evaluations/CLAUDE_CODE_EVAL.md` and `data/raw/ippl/WORKFLOW.md`. ### Step 1: Read WORKFLOW.md ```markdown file_path: data/...`

test_004 - How does IPPL profile MPI calls using mpiP?

- Verdict: ****Partial****
- Correct reference answer: IPPL profiles MPI calls with mpiP by preloading `libmpiP.so`, optionally setting `MPIP` flags, then parsing the produced `.mpiP` output with the bundled `scripts/mpiP.py`.
- Explanation: The answer overlaps with the Codex reference, but misses important scope, implementation detail, or exact terminology.
- Reference-keyword coverage: `0.35`
- Reference identifier/location hits: `2/5`
- Latency: `35.25s`
- Matched reference signals: `profiles, mpi, calls, mpip, MPIP, mpiP`
- Missing/weak reference signals: `preloading, libmpip.so, optionally, setting, flags, then, parsing, produced`
- Claude Code answer excerpt: `To understand how IPPL profiles MPI calls using `mpiP`, we need to explore the relevant files in the `data/raw/ippl` directory. Let's start by looking for any mentions of `mpiP` in the repository. ### Step 1: Search for `mpiP` Let's search through the files in `data/raw/ippl` to find any references to `mpiP`. ```bas...`

test_005 - What does the SLURM job template for Gwendolen in the README look like?

- Verdict: ****Correct****
- Correct reference answer: The README does include a Gwendolen SLURM template showing the `gmerlin6` cluster, `gwendolen` partition, GPU count, and `srun ./<your\_executable> <args> --kokkos-map-device-id-by=mpi\_rank`.
- Explanation: The answer captures the central facts of the Codex reference and includes the important technical nouns, symbols, or implementation locations.
- Reference-keyword coverage: `0.56`
- Reference identifier/location hits: `1/3`
- Latency: `14.08s`
- Matched reference signals: `readme, gwendolen, slurm, template, partition, gpu, count, srun`
- Missing/weak reference signals: `showing, gmerlin6, cluster, your\_executable, args, kokkos-map-device-id-by, mpi\_rank, srun ./<your\_executable> <args> --kokkos-map-device-id-by=mpi\_rank`
- Claude Code answer excerpt: `The README file does not explicitly contain a SLURM job template for Gwendolen. However, based on the context provided and the available files, I can infer that the relevant SLURM job script is `job.sh`, which is mentioned in the CLAUDE.md file. Here is the content of `job.sh`: ```bash #!/bin/bash #SBATCH --partitio...`